The FIFO Linking Conjecture: Affine Response and the Cut-Space Frontier

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Abstract

Let a finite graph be played by opening vertices into a FIFO queue and later closing the front. A close flips a bit exactly when the front has odd degree into the untouched set. The FIFO linking conjecture asserts that one isolated dummy lets either designated seat force even terminal parity on every graph.

We convert the conjecture into an exact affine problem. Every complete history has a disjointness vector in the $\mathbb{F}_2$ edge space of the complete graph. For a fixed attacker strategy, universal harmlessness is equivalent to zero lying in the affine hull of the compatible response vectors. At a front f, the edge space splits into the cut star at f and the edge space after deleting f; the quotient is the two-graph coboundary chart. The missing proof is therefore a causal, strategy-relative affine contraction across successive front levels.

We prove the transition and affine reductions, a parity-cell subclass, initial cut contraction, a two-bit handshake refinement, complete contraction against close-first attackers, the conditioned close-first tail, and the stopped defender-first root theorem. We also derive Eulerian curvature and response-factor formulations. Explicit counterexamples rule out childwise common-coset induction, bounded-support certificates, scalar Markov states, fixed-permutation contraction, and several natural pairing schemes. Exact minimax verifies the conjecture through eight real vertices plus the dummy, but no finite census is used as a universal argument. The remaining theorem is the construction of an odd multibranch response chain whose terminal moment is zero, or equivalently cut-valued in the Euler quotient.

1 The reduced game

Let $G = (R, E)$ be a finite simple graph on the *real* vertices, and adjoin one isolated vertex d. A state is (U, Q, ko), where U is the set of untouched vertices, Q is an ordered FIFO queue of open vertices, and ko records the one-move prohibition after a vertex is opened onto an empty queue. A legal move is

1. OPEN(x) for any $x \in U$, moving x to the back of Q; or
2. CLOSE(f) for the front f of Q, unless ko is set.

A close of f scores

$$\deg_U(f) \pmod 2.$$

If no move is legal, the forced pass only clears ko. This can occur only after U is empty, when no later close can score. The players have opposite targets for the parity of the total score.

Every vertex opens and closes exactly once. FIFO makes the opening and closing orders identical. Consequently the activity intervals $I_x = [o(x), c(x)]$ never properly nest. For an edge xy with

$o(x) < o(y)$, the edge contributes to the flip score precisely when $c(x) < o(y)$, i.e. precisely when I_x and I_y are disjoint. Thus the total score is

$$F_G(h) = \sum_{xy \in E(G)} [I_x \cap I_y = \varnothing] \pmod 2. \tag{1}$$

The proper-interval vocabulary is standard; the adversarial parity problem is not a standard queue-layout problem, because the queue stores vertices and the play itself constructs their intervals.

There is also an exact static encoding of the schedule which is useful for checking any proposed involution. Let $N = |R| + 1$, including the dummy, and let $\pi = (\pi_1, \dots, \pi_N)$ be the common opening and closing order. Put

$$r_b = \#\{\text{closes strictly before } O_{\pi_b}\}.$$

Proposition 1 (permutation–threshold normal form)**.** *Complete legal schedules are in bijection with pairs (π, r) in which π is a permutation and*

$$0 = r_1 \le r_2 \le \cdots \le r_N, \qquad 0 \le r_b \le b - 1,$$

subject to the ko condition

$$r_b = b - 1 \text{ and } b < N \quad \Longrightarrow \quad r_{b+1} = b - 1. \tag{2}$$

For $a < b$ one has

$$\pi_a \pi_b \in D(h) \quad \Longleftrightarrow \quad a \le r_b. \tag{3}$$

If $r_N = N - 1$, the unique forced pass occurs after O_{π_N} and before its close; otherwise the encoding is pass-free.

Proof. FIFO identifies the opening and closing permutations. The number of closes before successive opens is nondecreasing, and before the bth open at most $b - 1$ vertices have opened. Equality $r_b = b-1$ says that π_b opens onto an empty queue. Ko then forbids closing it before the next open, which is exactly (2). Conversely, before each O_{π_b} close the first $r_b - r_{b-1}$ queued vertices. The displayed bounds make these closes available, and (2) is precisely the only possible ko obstruction. After the last open, close the remaining queue, inserting the forced pass exactly in the stated singleton case. Finally, when $a < b$, I_{π_a} and I_{π_b} are disjoint exactly when π_a is among the r_b vertices already closed before O_{π_b}, proving (3). □

Corollary 2 (maximal-block score)**.** *Consider one maximal nonempty-queue block with opening order $v_1, \dots, v_b$, and let S be the vertices still untouched when the block ends. If t_i is the number of block vertices opened before C_{v_i}, then either $b = 1$, $S = \varnothing$, and the block is the terminal open–pass–close macro, or $b \ge 2$ and*

$$i + 1 \le t_i \le b \ (i < b), \qquad t_b = b, \qquad t_1 \le \cdots \le t_b,$$

and these conditions characterize maximal blocks. The charge of the ith close and the total block score are

$$c_i = \deg_S(v_i) + \sum_{j > t_i} [v_i v_j \in E(G)], \tag{4}$$

$$F_{\text{block}} = e_G(\{v_1, \dots, v_b\}, S) + \sum_{\substack{i<j \\ j > t_i}} [v_i v_j \in E(G)] \pmod 2. \tag{5}$$

Proof. The singleton case is exactly the ko-forced terminal macro. Assume $b \geq 2$. After C_{v_i} the queue has length $t_i - i$. It must remain positive for $i < b$ and become zero for $i = b$, giving the bounds; monotonicity records that opens accumulate as the FIFO closes proceed. Conversely, open the next vertices and perform C_{v_i} as soon as the count reaches t_i, resolving ties in FIFO order. The bounds make the queue positive until the last close, so this constructs the unique canonical block word. At C_{v_i} the untouched vertices are S together with exactly the later block vertices v_j for which $j > t_i$. This proves (4); summing over i gives (5). □

The block formula exposes a sharp boundary for the broad prevention/debt menu. A defender close cannot create debt under that menu: at debt zero only an even close is admitted, while at debt one an odd close discharges and an even close transports. Hence a block ends odd precisely when its last odd c_i belongs to the attacker; its final close always belongs to the defender, so necessarily $c_b = 0$. This condition is not by itself an exclusion. On the path $a - f - u$ plus isolated d, the completely legal line

$$O_a, O_d, C_a^{(1)}, C_d^{(0)}, O_f, O_u, C_f^{(0)}, C_u^{(0)}$$

ends with odd debt. The losing defender fork is the first reply O_d: the two-vertex block $\{a, d\}$ has score $\deg_{\{f,u\}}(a) = 1$. If an initial real opener has odd degree, handshaking instead supplies another odd real vertex; replying with it makes the immediate close–close branch even. This repairs the displayed fork but does not constrain longer block continuations.

2 The vector score

Write

$$\mathcal{E}_R = \mathbb{F}_2^{\binom{R}{2}}$$

for the vector space whose coordinates are unordered pairs of real vertices. For a completed history h, define its *disjointness vector*

$$D(h) = \sum_{\{x,y\} \subseteq R} [I_x \cap I_y = \varnothing], xy \in \mathcal{E}_R.$$

Pairing $D(h)$ with the adjacency vector A_G recovers (1).

There is an exact endpoint formula for the cut boundary of this score. It is useful because it isolates the part of any prospective local coefficient system which constant-coefficient topology cannot see. Temporarily put $V = R \sqcup \{d\}$ and let $\widehat{D}(h) \in E(K_V)$ include the dummy coordinates as well. Number only OPEN and CLOSE events, omitting the unique possible forced pass, and write $o_v, c_v \in \{1, \ldots, 2|V|\}$ for the two event positions of v.

Proposition 3 (endpoint-controller boundary)**.** *For every complete legal FIFO history and every $v \in V$,*

$$\deg_{\widehat{D}(h)}(v) = |V| + o_v + c_v \pmod 2. \tag{6}$$

Consequently, on the real edge space,

$$(\partial D(h))_v = |V| + o_v + c_v + \widehat{D}(h)_{vd}. \tag{7}$$

If $\bar{o}_v, \bar{c}_v$ are the positions of the real events after deleting O_d, C_d from the event word, whether or not the resulting word is legal, then equivalently

$$(\partial D(h))_v = |R| + \bar{o}_v + \bar{c}_v. \tag{8}$$

Thus the cut boundary records exactly whether the two endpoint events of each real interval are inherited by the same or opposite projected controllers.

Proof. Let v have opening rank b, let r earlier vertices have closed before O_v, and let t vertices have opened by C_v. FIFO makes v the bth vertex to close, so

$$o_v = b + r, \qquad c_v = b + t.$$

Exactly the r already-closed earlier vertices and the $|V| - t$ not-yet-opened later vertices have intervals disjoint from I_v. Hence

$$\deg_{\widehat{D}}(v) = r + |V| - t = |V| + o_v + c_v \pmod 2,$$

which proves (6). Deleting the dummy coordinate from $\widehat{D}$ gives (7).

Finally, deleting O_d, C_d changes the parity of an endpoint position exactly when that endpoint lies strictly between the two dummy events. Properness of the FIFO intervals says that exactly one endpoint of I_v lies there when I_v overlaps I_d, and zero modulo two lie there when the intervals are disjoint. Therefore

$$\bar{o}_v + \bar{c}_v = o_v + c_v + 1 + \widehat{D}_{vd}.$$

Since $|V| + 1 = |R|$ modulo two, substitution proves (8). □

At any moment let $L = U \cup Q$ be the live set. For a real $x \in L$, put

$$s_L(x) = \sum_{y \in (L \cap R) \setminus \{x\}} xy,$$

and set $s_L(d) = 0$. These are the vertex stars, or simplicial coboundaries, of the live complete graph.

Theorem 4 (vector live-star identity)**.** *Every completed history satisfies*

$$D(h) = \sum_{x \text{ opened}} s_{L_x}(x), \tag{9}$$

where L_x is the live set at the instant x opens.

Proof. Fix a real pair xy and suppose x opens first. The vertex y is live when x opens, so xy occurs once in $s_{L_x}(x)$. If the activity intervals overlap, x is still live when y opens and the coordinate occurs a second time in $s_{L_y}(y)$; the two occurrences cancel over $\mathbb{F}_2$. If the intervals are disjoint, x has already closed when y opens and there is no second occurrence. The coordinate of the right-hand side is therefore exactly the coordinate of $D(h)$. □

Corollary 5 (scalar live-degree identity)**.** *For every graph G,*

$$F_G(h) = \sum_{x \text{ opened}} \deg_{G[L_x]}(x) \pmod 2.$$

The threshold normal form does not permit a contraction one opening permutation at a time.

Proposition 6 (fixed-permutation contraction fails)**.** *There is a conditioned deterministic attacker strategy whose four response leaves have the same opening permutation but whose projected real disjointness vectors have affine hull excluding zero.*

Proof. Take real vertices a, b, c and dummy d. After the prefix O_a, O_d, put the attacker in the even positions. If the defender's third move is O_b, the attacker plays C_a; if it is C_a, play O_b. On move six, after the defender chooses O_c or C_d, play the other. The four resulting leaves are

$$B = O_aO_dO_bC_aO_cC_dC_bC_c, \qquad C = O_aO_dO_bC_aC_dO_cC_bC_c,$$
$$D = O_aO_dC_aO_bO_cC_dC_bC_c, \qquad E = O_aO_dC_aO_bC_dO_cC_bC_c.$$

They are legal single blocks, all with opening order (a, d, b, c). Their two free thresholds (r_3, r_4) are respectively $(0, 1), (0, 2), (1, 1), (1, 2)$, and direct use of (3) gives

$$D(B) = D(C) = ac, \qquad D(D) = D(E) = ab + ac.$$

The affine hull is $ac + \mathrm{span}\{ab\}$ and excludes zero. The unpruned threshold lattice has one further point $(0, 0)$ with zero moment; the attacker's move C_a prunes exactly that correction. Thus any global contraction must couple different opening-permutation branches. □

This upgrades the earlier scalar potential calculation to an identity before a graph is chosen. The graph enters only through the final linear functional $z \mapsto \langle A_G, z\rangle$.

3 Affine response is exactly the theorem

Fix which seat seeks odd parity, and fix a deterministic strategy S for that attacker. Move legality is graph-independent, so S can be regarded as a strategy in the schedule game before an adjacency vector is chosen. Let $\mathcal{H}_S$ be the set of terminal histories compatible with S and with arbitrary legal responses by the other player.

Theorem 7 (affine separation)**.** *The strategy S forces odd score for some graph on R if and only if*

$$0 \notin \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}.$$

Consequently, the general linking theorem is equivalent to the following graph-free statement for both choices of attacker seat:

$$\textit{for every } S, \qquad 0 \in \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}. \tag{10}$$

Proof. If S forces odd score for a graph with adjacency vector A, then $\langle A, D(h)\rangle = 1$ for every $h \in \mathcal{H}_S$. The whole affine hull therefore lies in the affine hyperplane $\langle A, z\rangle = 1$, which excludes zero.

Conversely, write the affine hull as $a + W$. If it excludes zero, then $a \notin W$. Elementary linear separation over $\mathbb{F}_2$ supplies a linear functional ℓ with $\ell(W) = 0$ and $\ell(a) = 1$. Every linear functional on $\mathcal{E}_R$ is pairing with the adjacency vector of a simple graph on R. For that graph, $\ell(D(h)) = 1$ for every response history, so S forces odd score. □

The separating graph can also be used to remove an apparent source of generality from a hypothetical counterexample. Say that a scalar policy is *positional* when its prescribed attacker move depends only on the full public state $(U, Q, \mathsf{ko}, \mathsf{turn}, g)$, including the accumulated score bit, and not on the history which reached that state.

Proposition 8 (positional scalar reduction)**.** *If* (10) *fails for either attacker seat, then it fails for a strategy which is positional after scalarization by one graph B. More precisely, a counterexample is equivalent to a finite set $\mathcal{W}$ of full public states, containing the initial state, such that every terminal state in $\mathcal{W}$ has score one, every defender state in $\mathcal{W}$ has all of its legal children in $\mathcal{W}$, and every attacker state in $\mathcal{W}$ has at least one legal child in $\mathcal{W}$.*

Proof. If affine contraction fails, Theorem 7 supplies a graph B for which the original history-dependent policy forces terminal score one. On the finite acyclic public-state graph, let $W(s)$ mean that the attacker can force terminal score one from s. Backward induction on the number of remaining legal events gives the Bellman alternatives: at an attacker state $W(s)$ holds exactly when it holds at some legal child, while at a defender state it holds exactly when it holds at every legal child. Choose one winning child once and for all at every winning attacker state. The resulting policy is positional and the set of its winning states is the required $\mathcal{W}$.

Conversely, choose one child in $\mathcal{W}$ at every attacker state. Closure at defender states makes every compatible terminal history have scalar score one. Pairing its disjointness vector with B is therefore one on every leaf, so their affine hull lies in the hyperplane $\langle B, z\rangle = 1$ and excludes zero. $\square$

Thus history-dependent pruning is not itself the essential obstruction. A proof may work with a Markov Bellman certificate, but it must retain the score sheet: two paths which reconverge after forgetting score can arrive on opposite sheets.

Equivalently, one must select an odd-cardinality formal multiset of response histories whose disjointness vectors XOR to zero. This is an $\mathbb{F}_2$-flow problem: attacker nodes route all incoming weight along the move prescribed by S, while defender nodes may split it among an odd number of outgoing moves. Uniformly summing all response leaves does *not* work in general; the flow must be selected.

There is a useful exact homogeneous recursion. At a history h, let $Z(h) \subseteq \mathbb{F}_2 \oplus \mathcal{E}_R$ be the span of $(1, D_h(\lambda))$ over response leaves λ below h, where D_h is the future disjointness increment. Give a close of a real front f the charge $c = \sum_{u \in U \cap R} fu$, and give an open, dummy close, or pass charge zero. This sums to $D(h)$ because a pair becomes disjoint exactly when its earlier vertex closes while the later one is untouched. Put

$$\tau_c(a, z) = (a, z + ac).$$

Then a terminal node has $Z = \operatorname{span}\{(1, 0)\}$; an attacker node, unary after fixing S, has $Z(h) = \tau_c Z(h')$; and a defender node has

$$Z(h) = \operatorname{span}_m \tau_{c_m} Z(h_m).$$

The desired affine response is exactly $(1, 0) \in Z(\text{root})$. Thus the open problem is a label-preserving contraction of this recursion, not an ordinary contraction of the response tree: ordinary boundaries have even augmentation, while the required leaf chain has odd augmentation.

The augmentation parity can be resolved explicitly rather than hidden inside the homogeneous span. This is the exact operation needed when an odd flow must split among several defender children.

Proposition 9 (parity-resolved continuation convolution)**.** *At a history h, with its residual history-dependent attacker policy retained, let Λ_h be the compatible terminal leaves and put*

$$\mathcal{C}_h^\epsilon = \{D_h(c) : c \in \mathbb{F}_2[\Lambda_h],\ \operatorname{aug}(c) = \epsilon\}, \qquad \epsilon \in \mathbb{F}_2.$$

Then $W_h := \mathcal{C}_h^0$ is a linear space and $A_h := \mathcal{C}_h^1$ is a nonempty affine W_h-coset. At a terminal node, $\mathcal{C}_h^0 = \mathcal{C}_h^1 = \{0\}$. If an attacker node uses its fixed move of charge c and has child h', then

$$\mathcal{C}_h^\epsilon = \epsilon c + \mathcal{C}_{h'}^\epsilon. \tag{11}$$

At a defender node with legal moves m, charges c_m, and children h_m,

$$\mathcal{C}_h^\epsilon = \bigcup_{\sum_m \epsilon_m = \epsilon} \sum_m \left(\epsilon_m c_m + \mathcal{C}_{h_m}^{\epsilon_m}\right). \tag{12}$$

Equivalently, choose $a_m \in A_{h_m}$, put $b_m = c_m + a_m$, and fix one reference move m_0. Then

$$W_h = \sum_m W_{h_m} + \operatorname{span}\{b_m + b_{m_0} : m \neq m_0\}, \tag{13}$$

$$A_h = b_{m_0} + W_h. \tag{14}$$

In particular a target t is feasible with odd augmentation exactly when $t + b_{m_0}$ belongs to the space on the right of (13)*.*

Proof. A formal leaf chain splits uniquely among the disjoint child leaf sets. Its augmentation and moment are respectively the sums of the child augmentations and of the translated child moments, giving (11) and (12). Two chains of the same augmentation differ by an even chain, so A_h is a W_h-coset. At a defender node, every odd child-parity profile is obtained from the profile supported on m_0 by toggling pairs of child parities. The even corrections inside each child give $\sum_m W_{h_m}$, and toggling m_0 together with m adds $b_{m_0} + b_m$. This proves (13) and (14). The rank

$$\rho(h) = 3|U| + |Q| + 1_{\mathsf{ko}}$$

strictly decreases under every OPEN, CLOSE, or forced pass, so the recursion is well founded. Its domain is the pair consisting of the history and the residual policy: two histories may reconverge to the same public (U, Q, ko) state while a history-dependent attacker retains different continuations. □

The same convolution can be flattened across an arbitrary decorated frontier; this is the exact bookkeeping needed below when prefixes have already accumulated different cut charges.

Corollary 10 (decorated-frontier decomposition)**.** *Let $\mathcal{F}$ be a finite antichain in the residual response tree which meets every terminal branch exactly once. For $q \in \mathcal{F}$, let Γ_q be the disjointness charge accumulated from the root to q, and write the odd continuation coset at q as*

$$A_q = a_q + W_q.$$

Then

$$A_{\mathrm{root}} = \operatorname{Aff}\{\Gamma_q + a_q : q \in \mathcal{F}\} + \sum_{q \in \mathcal{F}} W_q, \tag{15}$$

$$W_{\mathrm{root}} = \sum_{q \in \mathcal{F}} W_q + \operatorname{span}\{(\Gamma_q + a_q) + (\Gamma_{q_0} + a_{q_0}) : q \neq q_0\} \tag{16}$$

for any fixed $q_0 \in \mathcal{F}$.

Proof. Every root leaf lies below a unique $q \in \mathcal{F}$. Splitting a formal leaf chain by these disjoint descendant sets shows that its total moment is

$$\sum_q \big(\epsilon_q(\Gamma_q + a_q) + w_q\big), \qquad w_q \in W_q,$$

where ϵ_q is the augmentation of the q-part. The root chain is odd exactly when $\sum_q \epsilon_q = 1$. The odd sums of the decorated points $\Gamma_q + a_q$ form their affine hull, while the w_q vary independently; this proves (15). The even sums of the decorated points are generated by their pairwise differences from one reference point, which gives (16). □

The isolated dummy also gives an exact affine model of one OPEN fan, but only as long as the next attacker operator is common across its children.

Proposition 11 (virtual OPEN and the pruning obstruction)**.** *Let $U = R \sqcup \{d\}$ with d isolated, and let $\epsilon \in \mathbb{F}_2^U$ have odd augmentation. Its real virtual vertex*

$$\nu(\epsilon) = \sum_{r \in R} \epsilon_r r$$

determines ϵ uniquely, since $\epsilon_d = 1 + \sum_{r \in R} \epsilon_r$. If every selected OPEN child is followed by the same front close C_f, then the aggregate prefix charge is

$$\sum_{z \in U} \epsilon_z c_z = \sum_{r \in R} fr + f\nu(\epsilon). \tag{17}$$

This affine rule does not commute with a history-dependent attacker policy. On real vertices $0, 1, 2$ and dummy d, let the attacker first play O_0. After defender replies O_1 or O_2, it plays "OPEN the least untouched vertex, otherwise CLOSE"; after O_d, it plays "CLOSE if legal, otherwise OPEN the least untouched vertex". The three continuation affine sets, in coordinates $01, 02, 12$, are exactly

$$\mathcal{A}_1 = \mathcal{A}_2 = \{0\}, \qquad \mathcal{A}_d = 01 + 02 + \langle 12 \rangle. \tag{18}$$

Hence the nontrivial odd superposition of all three OPEN children has continuation coset

$$\mathcal{A}_1 + \mathcal{A}_2 + \mathcal{A}_d = 01 + 02 + \langle 12 \rangle,$$

which excludes zero despite the common zero OPEN charge.

Proof. Odd augmentation makes the displayed formula for ϵ_d immediate. After O_z, a common C_f has charge

$$c_z = \sum_{r \in R} fr + fz \quad (z \in R), \qquad c_d = \sum_{r \in R} fr.$$

Summing with coefficients ϵ_z proves (17).

For the policy obstruction, all compatible terminal histories are the following; thus the verification is a finite branch argument rather than a force-set computation:

	history	D_R
O_1	$O_0O_1O_2O_dC_0C_1C_2C_d$	0
	$O_0O_1O_2C_0O_dC_1C_2C_d$	0
O_2	$O_0O_2O_1O_dC_0C_2C_1C_d$	0
	$O_0O_2O_1C_0O_dC_2C_1C_d$	0
O_d	$O_0O_dC_0O_1C_dO_2C_1C_2$	$01 + 02$
	$O_0O_dC_0O_2C_dO_1C_2C_1$	$01 + 02$
	$O_0O_dC_0C_dO_1O_2C_1C_2$	$01 + 02$
	$O_0O_dC_0O_1C_dC_1O_2PC_2$	$01 + 02 + 12$
	$O_0O_dC_0O_2C_dC_2O_1PC_1$	$01 + 02 + 12.$

In a real child, after the attacker's next OPEN the defender can only OPEN the remaining dummy or close 0, giving the two listed rows. In the dummy child the forced C_0 leaves the defender the three choices O_1, O_2, C_d; the first two each split once more between an OPEN and the next front CLOSE, while the last creates the unique ko-forced row. This exhausts the policy tree. Direct interval inspection gives the last column and proves (18). □

The example is the smallest nontrivial odd OPEN superposition: with at most two children every odd subset is a singleton. It is not a root counterexample, because either singleton real branch already supplies the zero moment. What fails is precisely the attempt to replace the decorated family $z \mapsto a_z + W_z$ by one virtual vertex before applying the branch-dependent attacker continuations; symplectic Gram–Schmidt on the alternating adjacency form alone cannot make that replacement.

Sibling coupling is essential already at three real vertices. Let $d = 3$ and consider the defender fan after the attacker opens 0. One deterministic residual attacker has the three branch spaces

$$A_1 = 02 + \langle 12 \rangle, \qquad A_2 = 01 + \langle 12 \rangle, \qquad A_d = 01 + 02 + \langle 12 \rangle, \qquad W_1 = W_2 = W_d = \langle 12 \rangle. \quad (19)$$

The compatible schedules are replayed in `experiments/linking_game.py`. Direct interval calculation gives the displayed labels. Zero is absent from every individual A_m, even after all three even spaces are added, but the odd parity profile $(1, 1, 1)$ works because

$$02 + 01 + (01 + 02) = 0.$$

Thus no induction can send the odd target into one chosen defender child. This is minimal in real-vertex count: when the real edge space has dimension at most one, either the combined even space is the whole edge space, making a unit child feasible, or it is zero, in which case equal nonzero child offsets make every odd sum nonzero.

There is a useful dual form. Pair $\mathbb{F}_2 \oplus \mathcal{E}_R$ with itself by

$$\langle (\alpha, B), (a, z) \rangle = \alpha a + \langle B, z \rangle$$

and put $N_h = Z(h)^\perp$. For a move charge c, define the involution

$$\sigma_c(\alpha, B) = (\alpha + \langle B, c \rangle, B).$$

Proposition 12 (dual obstruction descent)**.** *The exact dual recursion is*

$$\begin{aligned}
&N_h = \{(0, B) : B \in \mathcal{E}_R\} && \textit{at a terminal node,}\\
&N_h = \sigma_c N_{h'} && \textit{at a fixed attacker move,}\\
&N_h = \bigcap_m \sigma_{c_m} N_{h_m} && \textit{at a defender node.}
\end{aligned}$$

Hence a counterexample is exactly a pair $(1, B) \in N_{\mathrm{root}}$.

More generally, let an odd formal flow from a node h reach a frontier $\mathcal{F}$ with coefficients ϵ_q and path charges Γ_q. If $(r_h, B) \in N_h$ and $(r_q, B) \in N_q$ are the inherited dual bits, then

$$\sum_{q \in \mathcal{F}} \epsilon_q r_q = r_h + \left\langle B, \sum_{q \in \mathcal{F}} \epsilon_q \Gamma_q \right\rangle. \quad (20)$$

Thus every odd frontier flow of actual B-charge zero from a bad node $r_h = 1$ has an odd number of bad exits.

For a fixed counterexample choose a bad reachable node lexicographically minimal in

$$\big(|(U \cup Q) \cap R|, \ \rho \big).$$

It is an attacker node, its prescribed move is an odd real CLOSE, and its child has dual bit zero. It cannot have a balanced ordered front pair.

Proof. Orthogonality against $\tau_c(a,z)=(a,z+ac)$ replaces α by $\alpha+\langle B,c\rangle$, proving the three dual recurrences. Along a selected path this gives $r_h=\langle B,\Gamma_q\rangle+r_q$; multiply by ϵ_q, sum, and use $\sum_q \epsilon_q=1$ to obtain (20).

At a bad defender node with $U\neq\varnothing$, every OPEN child has charge zero, preserves the number of live real vertices, and lowers ρ. If $U=\varnothing$, every CLOSE or pass has charge zero and gives the same contradiction. A lexicographically minimum bad node is therefore controlled by the attacker. Its prescribed child would be bad unless the move had charge one; only an odd real CLOSE can have that charge. Finally Theorem 35 supplies an odd frontier whose consecutive front closes have equal scalar charge and whose transported coefficient κ is B-isotropic. Equation (20) would then produce a lower-rank bad exit, excluding a balanced front at the minimum node. □

The predecessor of that minimum node has a much more rigid form.

Proposition 13 (minimum-bad predecessor normal form)**.** *Let h be lexicographically minimum as in Proposition 12. Write its prescribed move as the odd close C_x. Its immediate defender predecessor is exactly one of the following two types.*

1. *A CLOSE predecessor*

$$r_g=1+q_y \xrightarrow{C_y,\, q_y=\deg_U(y)} r_h=1 \xrightarrow{C_x,\, \deg_U(x)=1} r=0. \tag{A}$$

   Here the predecessor queue begins $(y,x,\ldots)$. If $q_y=1$, this is the good–bad–good odd–odd CLOSE spike. If $q_y=0$, then g is also bad and $|U_h|$ is necessarily even.

2. *A ko-protected domination state after the dummy has already been touched:*

$$r_g=1,\qquad Q_g=(x),\qquad \mathsf{ko}=1,\qquad \varnothing\neq U_g\subseteq R, \tag{B}$$

   where $|U_g|$ is even and $B(x,u)=1$ for every $u\in U_g$. The defender is forced to open some u; every such child is minimum bad, and the attacker immediately plays C_x with odd charge $|U_g|-1$.

   This second type has a forced one-step ancestry. Let j be the empty-queue attacker node whose prescribed O_x reaches g, and let e be the preceding defender node. Then

$$Q_e=(y),\qquad e\xrightarrow{C_y,\, \deg_{U_e}(y)=1} j \xrightarrow{O_x,\, 0} g, \tag{B$'$}$$

   and $r_e=0$. Thus (B), like the odd subcase of (A), is created from a good defender node by an odd singleton-front CLOSE; it merely inserts a zero-charge empty-queue OPEN before the minimum bad checkpoint.

In particular an untouched isolated dummy rules out every OPEN predecessor of a minimum bad node.

Proof. The node h is not the root, since its prescribed move is a CLOSE. A pass cannot precede it because $U_h\neq\varnothing$. Suppose first that the defender reached it by opening z at a node g. OPEN has zero charge, so g is bad. Every alternative OPEN $w\in U_g$ has a bad child with the same lexicographic rank as h. Proposition 12 forces the attacker at each such child to close the common real front x oddly. If $q=\deg_{U_g}(x)$, then

$$q+B(x,w)=1\qquad (w\in U_g). \tag{21}$$

If the dummy d were untouched, the equation for $w = d$ would give $q = 1$, whereas the equations for all real w would make every $B(x, w) = 0$ and hence $q = 0$, a contradiction.

Thus every vertex of U_g is real. Summing (21) over w gives

$$q = |U_g|(1 + q).$$

The cardinality $|U_g|$ is therefore even, $q = 0$, and $B(x, w) = 1$ for all $w \in U_g$. If ko were clear, the defender could instead play the even close C_x; its child would be bad with one fewer live real vertex. Hence ko is set, which at a reachable position forces $Q_g = (x)$. This is (B).

The remaining possibility is a defender close C_y from a queue beginning $(y, x, \ldots)$. Put $q_y = \deg_U(y)$. If $q_y = 0$, then g is bad and any defender OPEN child is bad. Moreover

$$\sum_{v \in U} \big(B(y, v) + B(x, v)\big) = q_y + \deg_U(x) = 1,$$

so the set of v which make the two front charges equal after O_v is odd. If $|U|$ were odd, each such child would have an even untouched set and a balanced front. Theorem 35 and Proposition 12 would then give a bad exit deleting y, x, with fewer live real vertices than h (whether or not $y = d$), a contradiction. Hence $q_y = 0$ forces $|U|$ even. If $q_y = 1$, the defender recurrence instead gives $r_g = 0$. Together with the odd C_x and its good child, these are exactly the two subcases of (A).

It remains to prove the ancestry assertion for (B). Since the dummy has already been touched and U_g is nonempty, the empty-queue predecessor j is not the root and cannot follow a pass. Its preceding move is therefore a defender close C_y from a ko-clear singleton queue at a node e. Put $A = \{x\} \sqcup U_g$ and $q_y = \deg_A(y)$. The fixed O_x has charge zero, so $r_j = r_g = 1$, while the defender recurrence at e gives

$$r_e = 1 + q_y.$$

Suppose $q_y = 0$. Then e is bad, and every alternative defender OPEN O_v, $v \in A$, leads to a bad attacker checkpoint with queue (y, v). In the even-order live graph on $\{y\} \sqcup A$, the full-degree parity class of y has even cardinality. After deleting y, it therefore supplies an odd nonempty set of vertices v for which the two front charges at (y, v) agree. The untouched set $A - \{v\}$ is even, so both η_{yv} fibres are even and Theorem 35 applies. Its odd zero-charge flow has a bad exit by Proposition 12; that exit deletes y, v. If y is real it starts with one more live real vertex than g and deletes two, while if $y = d$ it starts with the same number and deletes the one real vertex v. Either way the exit has one fewer live real vertex than g, hence also fewer than the chosen minimum node h. This contradicts the primary minimality of h. Hence $q_y = 1$ and $r_e = 0$, proving (B′). □

The ancestry statement cannot be strengthened by requiring one odd unit from every immediate OPEN child. The following counterexample is entirely strategy-theoretic; no computed force set is used.

Proposition 14 (the mandatory (B′) OPEN fan is false)**.** *Let*

$$y = 0, \qquad x = 1, \qquad A = \{2, 3, 4, 5\},$$

and use the primary and diagnostic graphs

$$B = \{01, 12, 13, 14, 15\}, \qquad L = \{02\}.$$

At the clear defender state

$$e : \qquad U = \{1, 2, 3, 4, 5\}, \qquad Q = (0), \qquad g_B = 0,$$

there is one history-dependent attacker policy below the six defender moves with the following constant terminal scores:

$$(B,L)(\mathcal{A}_1)=\{(0,1)\},\qquad (B,L)(\mathcal{A}_s)=\{(0,0)\}\quad (2\le s\le 5),\qquad (B,L)(\mathcal{A}_{C_0})=\{(0,1)\}.$$

Here $\mathcal{A}_s$ is the terminal continuation set below O_s, and $\mathcal{A}_{C_0}$ is the corresponding set below C_0. Consequently

$$L(\mathcal{A}_1+\mathcal{A}_2+\mathcal{A}_3+\mathcal{A}_4+\mathcal{A}_5)=\{1\},$$

so an odd flow constrained to use every immediate OPEN face cannot contract.

Proof. We first record the elementary star strategy used in every branch. Once the center 1 is open, let k be the number of queued leaves ahead of it and let m be the number of untouched leaves. The only remaining B-charge is the charge m of C_1. At an attacker turn, play

$$\begin{cases} C_1, & k=0, m \text{ even},\\ \text{OPEN an untouched leaf}, & k=0, m \text{ odd}, \text{ or } k>0, m>0,\\ \text{CLOSE the front}, & k>0, m=0. \end{cases}$$

This forces an even future B-charge from every attacker turn, and from every defender turn satisfying $k>0$ or m even. Indeed, mutual induction on $k+m$ is immediate: a defender OPEN gives the attacker $(k,m-1)$, a defender prefront CLOSE gives $(k-1,m)$, and at $k=0$ a defender C_1 is already even when m is even. Each prescribed attacker move either finishes evenly or strictly decreases the same rank. Fix the least-labelled leaf whenever the rule asks for an OPEN.

Now consider the six first moves at e. After O_1, the attacker plays C_0, of charge $(B,L)=(0,1)$, and invokes the star strategy at the defender state $(k,m)=(0,4)$. After O_2, it plays O_1; the intervals of $0,2$ already overlap, and the star strategy begins at the defender state $(k,m)=(2,3)$. These give respectively the first two displayed score sets.

Let next $s\in\{3,4,5\}$ and write $\{a,b\}=A-\{2,s\}$. The attacker first plays O_2, fixing the L-score at zero. If the defender now opens 1, use the star strategy; if it opens a or b, open 1 and then use that strategy. If instead the defender closes 0, that close has B-charge one; answer by C_s, also of B-charge one. From the resulting defender state, an OPEN is answered by opening 1 if necessary and then using the star strategy. The only other move is C_2, which supplies a third unit B-charge and leaves the attacker at an empty queue with $U=\{1,a,b\}$. Play O_1; ko forces the defender to open a or b, and C_1 then sees the one remaining untouched leaf and supplies the fourth unit. All later B- and L-charges vanish. Thus every O_s branch has total $(B,L)=(0,0)$.

Finally, the defender move C_0 itself has charge $(1,1)$. The attacker opens 1 at the empty queue, ko forces the defender to open one of $2,3,4,5$, and the attacker closes 1 with B-charge three and L-charge zero. No later edge can score, so this branch totals $(0,1)$. The policies on the six disjoint child trees combine into the asserted single history-dependent policy. □

This refutes only the mandatory all-five transversal, not the global affine contraction. Under the same policy every face $s=2,3,4,5$ contains a literal zero-moment leaf; compatible continuations are

$$\begin{array}{c|l} 2 & O_2O_1O_3O_4O_5C_0C_2C_1C_3C_4C_5\\ 3 & O_3O_2O_1O_4O_5C_0C_3C_2C_1C_4C_5\\ 4 & O_4O_2O_1O_3O_5C_0C_4C_2C_1C_3C_5\\ 5 & O_5O_2O_1O_3O_4C_0C_5C_2C_1C_3C_4. \end{array}$$

All OPENs precede all CLOSEs, so their disjointness moment is zero. A one-face odd flow therefore already contracts; the example supplies no odd-winning strategy and no root counterexample.

The minimum-bad hypotheses do impose more structure on the first response, but that structure still has one exact obstruction.

Proposition 15 (the (B′) first-response forest criterion)**.** *At a (B′) predecessor put*

$$A = N \sqcup Z, \qquad N = \{v : B(y,v) = 1\}, \qquad Z = \{v : B(y,v) = 0\}.$$

Thus $|A|$ and $|N|$ are odd and $|Z|$ is even. Mark a vertex $v \in N$ as grounded when the attacker answers O_v by the immediate zero close C_y. For every other canonical branch, suppose its first attacker response is O_{w_v} and the defender then plays C_y. Minimum badness forces v, w_v to have opposite y-colours. Form the bipartite multigraph H on A with edges $\{v, w_v\}$, and regard the grounded vertices as marked components.

The resulting canonical good C_y exits contain an odd chain whose prefix decoration is zero if and only if every component of H containing no grounded vertex has even order.

Proof. Write $p = \sum_{a \in A} a$. The prefix decorations are

$$\Gamma_v = y(p+v) \quad (v \text{ grounded}), \qquad \Gamma_v = y(p+v+w_v) \quad (v \text{ an edge branch}).$$

For branch coefficients α_v, their sum is zero exactly when

$$p = \sum_{v\ mgrounded} \alpha_v v + \sum_{v\ medge} \alpha_v (v + w_v). \tag{22}$$

Conversely, any solution of (22) already has odd branch augmentation. Taking total coordinate augmentation says that an odd number of grounded units is selected, because $|A|$ is odd. Summing only the N-coordinates then says that an even number of edge branches is selected, because every edge crosses $N \mid Z$ and $|N|$ is odd.

Inside one connected component, its edge-incidence vectors span exactly the even-weight subspace. A grounded unit enlarges this to the full coordinate space of that component. Hence the all-one vector p is representable precisely when every component without such a unit has even order, proving the criterion. □

The criterion is not automatic from (B′). Take

$$A = \{x, z, z'\}, \qquad E(B) = \{yx, xz, xz'\},$$

with no grounded branch and first responses

$$x \longmapsto z, qquadz \longmapsto x, qquadz' \longmapsto x.$$

Here $N = \{x\}$, $Z = \{z, z'\}$, and x dominates the even set $\{z, z'\}$ exactly as required in (B′). The response graph is one ungrounded component of odd order three, while the three decorations are

$$\Gamma_x = yz', \qquad \Gamma_z = yz', \qquad \Gamma_{z'} = yz.$$

Only the even pair $\Gamma_x + \Gamma_z$ cancels. Thus even an alternating two-cycle may carry an odd in-tree defect. This is a no-go for a proof using only the first-response map: descendant continuation spaces and the other defender replies remain available, so it is not a counterexample to the linking theorem.

This is a genuine monotone reduction, but not yet the theorem. Case (A) cannot be extended backwards into a corridor of odd CLOSE pairs by local reasoning. For example, take $U = \{a, d\}$,

$Q = (z, y, x)$, with d isolated and the only displayed separator edges xa, ya. There is a residual policy with the exact dual pattern

$$0 \xrightarrow{C_z,0} 0 \xrightarrow{C_y,1} 1 \xrightarrow{C_x,1} 0:$$

after the defender alternatives O_a and O_d, the attacker can respectively use the now-even C_y and the zero OPEN O_a, and all later charges vanish. Thus the unresolved global task has two sharply related terminal patterns. The odd subcase of (A) and all of (B) begin at a good defender singleton whose odd close creates badness; (B) then carries that bad bit through one empty-queue OPEN before the ko-protected domination fan. The remaining even subcase of (A) can occur only with an even untouched set. One must exclude these patterns by coupling earlier bad siblings; neither endpoint can be discarded from its public state alone.

There is a sharper target when the separating graph is Eulerian. It removes exactly the directions which no Euler adjacency vector can see.

Theorem 16 (Euler quotient separation)**.** *Fix a deterministic attacker strategy S on a vertex set V, and write its response affine set as*

$$\mathcal{A}_S = \operatorname{Aff}\{D(h) : h \in \mathcal{H}_S\} = a + W \subseteq E(K_V).$$

There is no Eulerian graph on V for which S forces odd score if and only if

$$\mathcal{A}_S \cap \operatorname{Cut}(V) \neq \varnothing, \qquad \textit{equivalently} \qquad a \in W + \operatorname{Cut}(V). \tag{23}$$

After choosing any front f, this is equivalently

$$0 \in \operatorname{Aff}\{T_f D(h) : h \in \mathcal{H}_S\}. \tag{24}$$

Proof. Under the edge-space dot product, Eulerian graphs are the cycle space $Z(K_V)$, and

$$Z(K_V) = \operatorname{Cut}(V)^{\perp}.$$

If (23) holds, an Euler adjacency vector cannot pair to one on every response score. Conversely, if $a \notin W + \operatorname{Cut}(V)$, binary linear separation supplies

$$A \in (W + \operatorname{Cut}(V))^{\perp} \subseteq Z(K_V), \qquad \langle A, a\rangle = 1.$$

This A is the adjacency vector of an Eulerian simple graph, annihilates W, and therefore pairs to one on all of $a + W$. The final equivalence follows from Theorem 18, whose kernel is exactly $\operatorname{Cut}(V)$. $\square$

Thus the Euler kernel does not require the graph-free target $0 \in \mathcal{A}_S$: a cut-valued odd response flow is enough. This distinction is substantial. For even $|V|$ the bicycle space $Z(K_V) \cap \operatorname{Cut}(V)$ has dimension $|V| - 2$; demanding that response differences span those invisible directions is strictly stronger than the game. The remaining problem is the quotient contraction (24), not full edge-space spanning.

Computational evidence. The existing exhaustive graph census through eight real vertices plus dummy, both attacker seats, implies (10) in those dimensions by Theorem 7. Independent graph-free policy searches also found no affine offset through eight real vertices; selected structured policies at nine and ten real vertices likewise had zero offset. These are finite checks, not a proof of (10).

4 A solved parity-cell subclass

The true/false-twin argument in the earlier draft can be weakened substantially.

Theorem 17 (parity-cell pairing). *Suppose all vertices, including the dummy if present, admit a perfect partition $\mathcal{M}$ into two-element cells such that*

$$e_G(A, B) = 0 \pmod 2 \qquad \text{for every distinct } A, B \in \mathcal{M}. \tag{25}$$

Then the second player forces even flip parity.

Proof. When the first player opens one member of a cell A, the second opens its mate. When the first player closes the FIFO-front member of A, its mate is the next queue front and the second closes it. Thus untouched vertices are always a union of whole cells, and the queue is a concatenation of whole cells. Ko causes no exception because the forced second opening completes the first cell.

When the two members a, a' of A close consecutively, their two flip bits sum to

$$\deg_U(a) + \deg_U(a') = \sum_{B \subseteq U} e_G(A, B) = 0$$

by (25). Every flip is contained in one such consecutive pair, so the total parity is even. □

Twin pairs satisfy (25), but the condition only asks for even cross-parity between cells; no automorphism is required.

Computational evidence. This theorem is still not general. Among the 1,044 isomorphism classes on seven real vertices, 64 boards obtained by adjoining the isolated dummy have no such partition. The first witness in the local ordering has real edges

$$02, 04, 06, 14, 15, 23.$$

The general linking theorem nevertheless passes exact minimax on every one of those boards. Seidel switching cannot repair the pairing obstruction: between two two-cells A, B, switching a set S toggles

$$|A \cap S||B \setminus S| + |A \setminus S||B \cap S| \equiv ab + ab = 0 \pmod 2.$$

5 FIFO gives a cut-space filtration

Let $L \subseteq R$ be a live real set and let $f \in L$ be the FIFO front. Let $E(L) = \mathbb{F}_2^{\binom{L}{2}}$ and let

$$\mathrm{Cut}(L) = \mathrm{span}\{s_L(x) : x \in L\}.$$

This is the cut space of the complete graph on L.

Theorem 18 (front splitting). *There is a canonical direct sum*

$$E(L) = \mathrm{Cut}(L) \oplus E(L \setminus \{f\}). \tag{26}$$

The projection to the second summand is

$$(T_f z)_{ij} = z_{ij} + z_{fi} + z_{fj}, \qquad i, j \neq f. \tag{27}$$

It kills every live star and is the identity on the canonical $E(L \setminus \{f\})$ summand.

Proof. For $x = f$, the two incident coordinates in (27) cancel; for $x = i \neq f$, the ij and fi coordinates of $s_L(i)$ cancel. Hence T_f kills all generators of $\mathrm{Cut}(L)$. If z is supported away from f, then $T_f z = z$. Finally,

$$\dim \mathrm{Cut}(L) + \dim E(L \setminus \{f\}) = (|L| - 1) + \binom{|L|-1}{2} = \binom{|L|}{2},$$

so the two subspaces are complementary and exhaust $E(L)$. □

There is a front-independent description of the same quotient. Regard $z \in E(L)$ as a simplicial 1-cochain on the full simplex on L, and put

$$(\delta z)_{ijk} = z_{ij} + z_{ik} + z_{jk}.$$

Then $\ker \delta = \mathrm{Cut}(L)$ and

$$(T_f z)_{ij} = (\delta z)_{fij}. \tag{28}$$

Thus T_f is the chart based at f of the odd-triple two-graph of z; graphically it is Seidel complementation at f followed by deletion of f [4, 5]. Equivalently, if B is the alternating form with Gram matrix z and

$$E_0 = \{x \in \mathbb{F}_2^L : \textstyle\sum_i x_i = 0\}, \qquad b_i^{(f)} = e_i + e_f,$$

then $T_f z$ is the Gram matrix of $B|_{E_0}$ in the basis $\{b_i^{(f)} : i \neq f\}$. Changing the basepoint from f to g gives

$$b_f^{(g)} = b_g^{(f)}, \qquad b_i^{(g)} = b_i^{(f)} + b_g^{(f)}.$$

In particular, after restricting away from distinct f, g,

$$T_g(T_f z) = T_g z. \tag{29}$$

This absorption law says that earlier quotient gauges cancel exactly; it does not say that their affine payoff offsets cancel.

While f remains front, opens do not change the live set. By Theorem 4, all their contributions lie in the current $\mathrm{Cut}(L)$ layer. Once FIFO deletes the real front f, every later contribution is supported on $E(L \setminus \{f\})$. Repeating the splitting along the forced closure order gives a triangular decomposition of the complete play score. This is the missing structural use of FIFO: the earlier scalar potential was valid even if any queued vertex could close and therefore could not prove the theorem.

The dummy needs a companion state in this induction. Its star is zero, but closing it does not shrink the real edge space. A proof must use that zero star to adjust affine augmentation and then transfer the odd response flow to the unchanged real live set. Treating the dummy as an ordinary deleted front silently loses precisely the exceptional case.

6 What the cone homotopy does and does not contract

Fix a real front f and a terminal continuation λ. Let $p_f(\lambda) \in \mathbb{F}_2^{L\setminus\{f\}}$ be the indicator of the real vertices still untouched when f closes, and let $W_f(\lambda)$ be the away-f disjointness vector accumulated by the whole continuation. Directly from the definition of disjoint intervals,

$$D_h(\lambda) = \sum_i p_f(\lambda)_i \, fi + W_f(\lambda), \qquad T_f D_h(\lambda) = W_f(\lambda) + \delta p_f(\lambda), \tag{30}$$

where $(\delta p)_{ij} = p_i + p_j$. Hence for every formal response-leaf chain c, of either augmentation,

$$D_h(c) = 0 \iff p_f(c) = 0 \text{ and } W_f(c) = 0. \tag{31}$$

The two-graph chart therefore replaces the desired edge moment by a *joint* cut/continuation moment; it does not discard the correlation.

One half of that joint target does contract cleanly at the root.

Lemma 19 (initial cut-moment contraction)**.** *Suppose the attacker opens a real initial front f, so the isolated dummy remains untouched and ko forbids the defender's immediate close. For every fixed attacker strategy up to the close of f,*

$$0 \in \operatorname{Aff}\{p_f(\lambda) : \lambda \textit{ is a compatible pre-close response}\}.$$

Proof. If zero were absent, separation would give weights $a_i \in \mathbb{F}_2$ on the remaining real vertices such that the attacker forced $a \cdot p_f = 1$. Give the dummy weight zero, and let s be the total weight of the still-unselected vertices. On the ko-forced first response, the defender selects the dummy when $s = 0$, and selects a weight-one real vertex when $s = 1$; either choice leaves $s = 0$. On every later response, close f when $s = 0$, and select a weight-one vertex when $s = 1$. If the attacker selects weight zero, s remains zero and the defender closes. If the attacker selects weight one from a zero total, the number of remaining weight-one vertices was positive and even, so another remains and the defender restores $s = 0$. Whenever the attacker closes f, therefore, $a \cdot p_f = 0$, a contradiction. □

The exact residue after this contraction is a Schur complement, rather than an automatically exact spectral-sequence page. Let $V_h = \mathbb{F}_2[\Lambda_h]$ at a fixed real front f, extend p_f and W_f linearly, and suppose $c_0 \in V_h$ has

$$\operatorname{aug}(c_0) = 1, \qquad p_f(c_0) = 0.$$

Put $K_f = \ker(\operatorname{aug}, p_f)$. Then

$$\omega_f(S) = [W_f(c_0)] \in E(L - \{f\})/W_f(K_f) \tag{32}$$

is independent of the chosen c_0, and vanishes exactly when c_0 can be corrected to an odd zero-moment chain. Indeed, any two choices differ by K_f, while (31) says that an admissible correction $k \in K_f$ must solve $W_f(k) = W_f(c_0)$. This is the honest first filtered obstruction; attacker pruning gives no chain complex whose ambient simplicial $\delta^2 = 0$ would make it vanish automatically.

The first phase-pivot differential is visible in one commuting real diamond. If x is front and z untouched, the two orders OPEN–CLOSE and CLOSE–OPEN have charge vectors

$$\Gamma_{OC} = s_x(U - \{z\}), \qquad \Gamma_{CO} = s_x(U), \qquad \Gamma_{OC} + \Gamma_{CO} = xz.$$

Their p_x values differ by e_z, and $T_x(xz) = \delta e_z$ is the cut star of z on the residual live set. Successive matching shifts therefore create rooted-triangle holonomy.

Two exact four-real-vertex witnesses delimit this filtration. First, the compatible pair

$$h_1 = O_0 O_1 O_d C_0 O_2 C_1 O_3 C_d C_2 C_3,$$
$$h_2 = O_0 O_d O_1 C_0 O_3 C_d C_1 C_3 O_2 P C_2$$

has equal $p_0 = e_2 + e_3$, but

$$D(h_1) = 02 + 03 + 13, \qquad D(h_2) = 02 + 03 + 12 + 23, \qquad W_0(h_1 + h_2) = 12 + 13 + 23.$$

Thus the first surviving class may be the genuine triangle rather than a cut. This order is minimal because after deleting f a non-cut class requires at least three real vertices.

More sharply, one deterministic policy admits three leaves

$$
\begin{aligned}
g_1 &= O_0O_1O_2O_dC_0O_3C_1C_2C_dC_3,\\
g_2 &= O_0O_2O_3O_dC_0O_1C_2C_3C_dC_1,\\
g_3 &= O_0O_dO_2C_0O_1C_dC_2C_1O_3PC_3.
\end{aligned}
$$

Their respective (p_0, W_0) values are

$$(e_3, 0), \qquad (e_1, 0), \qquad (e_1 + e_3, 13 + 23).$$

The odd sum cancels p_0, and its residual $13 + 23$ pairs trivially with the unique residual triangle $12 + 13 + 23$, so both the first cut and the triangle quotient vanish; nevertheless its full moment is the nonzero next-front cut $13 + 23$. The three histories occupy different residual phases, so the initial cut contraction cannot simply be reapplied to their sum as one child state. Vanishing of two ambient filtration layers is therefore a strict false positive; the parity-resolved convolution must couple further siblings across levels.

The scalar separation proof has a useful two-round refinement. It repairs the first explicit failure of degree-only pairing, although it still does not iterate through an arbitrary FIFO prefix.

For a finite graph H, put

$$p_v = \deg_H(v) \pmod 2, \qquad b_v = \sum_{u \sim_H v} p_u \pmod 2, \qquad c(v) = (p_v, b_v).$$

Thus b_v is the parity of the odd-degree neighbours of v.

Lemma 20 (two-bit handshake refinement)**.** *If H has even order, every one of the four colour classes $c^{-1}(i,j)$ has even cardinality. In particular every vertex has a distinct same-colour mate.*

If $c(v) = c(w)$ and $A = \{v, w\}$, put $r_x = e_H(x, A) \pmod 2$ for $x \notin A$. Then each of the four sets

$$\{x \notin A : (p_x, r_x) = (i, j)\}, \qquad (i, j) \in \mathbb{F}_2^2,$$

also has even cardinality.

Proof. Let $O = \{v : p_v = 1\}$ and $E = V(H) \setminus O$. Handshaking gives $|O| = 0$ in $\mathbb{F}_2$. Moreover

$$\sum_{v \in O} b_v = 2e_H(O) = 0,$$

so both b-fibres in O are even. The cut $e_H(O, E)$ is even, since

$$e_H(O, E) = \sum_{v \in O} \deg_H(v) = |O| = 0$$

in $\mathbb{F}_2$. Hence the $b = 1$ fibre in E is even; $|E|$ is even because $|V(H)|$ and $|O|$ are even, so the remaining fibre is even as well.

For the second assertion, the four parity moments of the map $x \mapsto (p_x, r_x)$ on $V(H) \setminus A$ vanish:

$$\sum 1 = 0, \quad \sum p_x = 0, \quad \sum r_x = p_v + p_w = 0, \quad \sum p_x r_x = b_v + b_w = 0.$$

The internal edge vw, when present, contributes $p_v + p_w = 0$ to the last two displayed identities. The four fibre indicators are the four polynomials $(1+p)(1+r), (1+p)r, p(1+r), pr$ over $\mathbb{F}_2$; their sums therefore vanish. $\square$

Remark 21 (what the second bit buys). Suppose an attacker opens v and the defender opens a same-colour mate w. If the attacker closes v, closing w makes the two flip charges cancel. If the attacker instead opens x, Lemma 20 supplies a y with both $p_y = p_x$ and $e_H(y, A) = e_H(x, A)$. Thus the new cell matches the opening-charge parity and is orthogonal to the first FIFO cell. The equality $b_v = b_w$ is exactly the missing joint moment in the first degree-only counterexample.

This is a two-round statement, not an induction: the new cell can split the four balanced fibres unevenly. A later response may have to close rather than open, which is the correlated continuation obstruction in Question 40.

Computational evidence. Exact minimax on every graph isomorphism class through seven real vertices plus the dummy gives a stronger root fact: whenever the same-colour mate of Lemma 20 applies, *every* such reply is winning for the second-seat even player. When the real graph has even order and the attacker opens the unmatched dummy, every real second opening is winning through six real vertices. The first-seat even player also wins by opening the dummy throughout the seven-vertex census. The optional `refinement` stage of `experiments/linking_game.py` reproduces these checks. They support the two-bit prefix but do not prove that its balanced fibres survive later FIFO switches.

7 Scalar response faces and the least counterexample

It is useful to isolate exactly what an induction on a smaller even board would have to prove. Fix the player seeking odd parity and, for any game state h, let $R(h) \subseteq \mathbb{F}_2$ be the set of future charge bits that this attacker can force. If a move of charge c leads to h', write $c + R(h') = \{c + r : r \in R(h')\}$. Backward induction gives

$$R(h) = \begin{cases} \bigcup_m (c_m + R(h_m)), & h \text{ is an attacker node,} \\ \bigcap_m (c_m + R(h_m)), & h \text{ is a defender node,} \end{cases} \qquad R(\text{terminal}) = \{0\}. \tag{33}$$

Here a forced pass is the sole move from its state and has charge zero. For a live vertex set A, use the following canonical roots:

	Q	U	mover / ko
$T(A)$	$()$	A	attacker / false
$K_x(A)$	(x)	$A - \{x\}$	defender / true
$S_x(A)$	(x)	$A - \{x\}$	defender / false
$Q_{xy}(A)$	(x, y)	$A - \{x, y\}$	attacker / false.

The same letters denote their force sets. The boundary values are $T(\varnothing) = K_x(\{x\}) = \{0\}$. Away from the empty-union boundary, directly from (33),

$$T(A) = \bigcup_{x \in A} K_x(A) \quad (A \neq \varnothing), \qquad K_x(A) = \bigcap_{y \neq x} Q_{xy}(A) \quad (|A| \geq 2), \tag{34}$$

and

$$S_x(A) = K_x(A) \cap \left(\deg_{A-\{x\}}(x) + T(A - \{x\})\right). \tag{35}$$

Thus the scalar even-board statement is

$$E(A) : \qquad 1 \notin T(A) \quad (|A| \text{ even}). \tag{36}$$

It says that a first-seat odd attacker cannot force odd parity; by finite determinacy, the second player can force even parity.

The four possible force sets have an exact Boolean form. Write $r_h(t) = 1_{t\in R(h)} \in \mathbb{F}_2$. Then

$$r_h(t) = \begin{cases} 1 + \prod_m \bigl(1 + r_{h_m}(t + c_m)\bigr), & h \text{ an attacker node,} \\ \prod_m r_{h_m}(t + c_m), & h \text{ a defender node.} \end{cases} \tag{37}$$

In particular

$$K_x(t) = \prod_{y \neq x} Q_{xy}(t), \qquad T(t) = 1 + \prod_x (1 + K_x(t)), \qquad S_x(t) = K_x(t) T_{A-x}\bigl(t + \deg_{A-x}(x)\bigr). \tag{38}$$

Proposition 22 (the universal-full-fan residual)**.** *For an even graph H, put*

$$P(H): \qquad 0 \in S_z^H(V) \quad (z \in V).$$

Then

$$P(H) \iff \begin{cases} 0 \in Q_{zy}(H) & (z \neq y), \\ \deg_H(z) \in T(H - z) & (z \in V). \end{cases}$$

If $P(H)$ and $1 \in T(H)$, there is a vertex x such that

$$Q_{xy}(H) = \mathbb{F}_2 \qquad (y \neq x). \tag{39}$$

Consequently $P(H) \Rightarrow T(H) = \{0\}$ is equivalent to excluding the universal full row (39)*. If, in addition, H has an even-cross perfect pairing as in Theorem 17, then $T(H) = \{0\}$.*

Proof. Equations (35) and (34) give the first equivalence. If $1 \in T(H)$, some K_x contains 1. Under $P(H)$ the same K_x also contains zero, so it equals $\mathbb{F}_2$; its intersection formula forces every Q_{xy} in that row to equal $\mathbb{F}_2$. Conversely such a row gives $1 \in K_x \subseteq T$. Finally the parity-cell strategy forces even score from the empty-queue second-seat root, so $1 \notin T(H)$, while $P(H)$ already gives $0 \in T(H)$. □

Computational evidence. The implication $P(H) \Rightarrow T(H) = \{0\}$ holds on the complete graph census through order eight; the numbers of P-classes in even orders $2, 4, 6, 8$ are respectively $1, 4, 18, 262$, and every one admits an even-cross pairing. This is not yet a proof. The most obvious Boolean parity repairs already fail: on the path $01, 03, 12$, all $S_z = \{0\}$ but the row at 0 is $\{0\}, \{0\}, \mathbb{F}_2$; even the same-degree subrow can have odd full-set count at order six. Nor does Eulerianity alone supply the static certificate: an exact order-fourteen Eulerian graph has no even-cross perfect pairing. Thus the missing step is a joint identity internal to the ordered Q-subtrees, not a rowwise Möbius parity or a static pairing induction.

Lemma 23 (least-counterexample odd routing)**.** *Suppose A is a least-order even counterexample to* (36)*. Then there is an $x \in A$ with $1 \in K_x(A)$. Put*

$$Y = \{y \in A - \{x\} : \deg_A(y) = \deg_A(x) \pmod 2\}.$$

The set Y has odd cardinality. For every $y \in Y$, every target-1 strategy witnessing $1 \in Q_{xy}(A)$ begins by opening a third vertex z_y; it cannot begin by closing x.

At the resulting defender state (x, y, z_y), such a strategy must satisfy both

$$1 + \deg_{A-\{x,y,z_y\}}(x) \in Q_{yz_y}(A - \{x\}) \tag{40}$$

and target-1 forceability after every defender opening $w \in A - \{x, y, z_y\}$.

Proof. Direct play proves $E(A)$ for $|A| = 0, 2$: on two vertices the forced opening pair is followed by two zero-charge closes. Hence a least counterexample has order at least four, and Equation (34) supplies x without meeting either boundary case. Each degree-parity class in an even-order graph has even cardinality, so deleting x leaves the odd set Y.

Fix $y \in Y$. If the attacker closes x from $Q_{xy}(A)$, the charge is

$$c = \deg_{A-\{x,y\}}(x),$$

and the child is $S_y(A - \{x\})$. By minimality, $T(A - \{x, y\}) \subseteq \{0\}$, so (35) gives

$$S_y(A - \{x\}) \subseteq \{\deg_{A-\{x,y\}}(y)\}.$$

But equality of the full degree parities gives

$$\deg_{A-\{x,y\}}(x) = \deg_{A-\{x,y\}}(y),$$

where the possible edge xy cancels from both sides. The remaining target after the close is $1+c$, not c, so this branch cannot force target 1. The attacker must therefore open some z_y. The defender may now close x or open any remaining w. Applying (33) to those children gives (40) and the stated universal open conditions. □

The odd routing is a real strengthening over the handshake lemma: it couples an odd family of sibling histories. It still cannot be collapsed one sibling at a time. A tempting twisted induction would label $C = A - \{x, y\}$ by $a(u) = B(x + y, u)$ and ask for a w with

$$1 + a(z) + a(w) \notin Q_{zw}(C).$$

Computational evidence. This is false first at $|C| = 4$. Take the path 5–2–3–4, put $(a_2, a_3, a_4, a_5) = (1, 0, 1, 0)$, and $z = 5$. Exact recursion gives

$$T(C) = \{0\}, \qquad Q_{52} = \{0\}, \quad Q_{53} = \{0, 1\}, \quad Q_{54} = \{0\},$$

while the three shifted targets are $0, 1, 0$. All three survive. This is realized inside the six-vertex graph

$$E = \{03, 12, 13, 14, 23, 25, 34\}, \qquad (x, y) = (0, 1).$$

Hence ordinary smaller-board induction, even augmented by one complete cut bit, does not prove Lemma 23's required joint sibling cancellation.

The lemma cannot simply be followed by an independent quotient induction. On real vertices $0, 1, 2$ plus d, fix a second-seat attacker so that the line

$$O_d, O_1, C_d, O_0, C_1, O_2, C_0, C_2$$

is forced below the displayed defender choices. At the C_1 branch from $Q = (1, 0), U = \{2\}$, one has $p_1 = e_2$, $W_1 = 0$, $D = 12$, and $T_1D = 02$. The subtree below that branch has one leaf. The fictitious 02 terms in $W + \delta p$ cancel only after the cut and quotient pieces are recombined; no compatible continuation chain realizes either term separately. This is the smallest broken commuting diamond behind the false branchwise induction.

8 Obstructions to deterministic local induction

The live-degree pairing heuristic is correct only conditionally. At a turn of the block initiator the queue has even length. If the initiator opens x, the responder would like to open an untouched y with the matching live-degree parity. If no such y exists and the old queue is nonempty, closing the old front rotates

$$(a_1, b_1, \dots, a_r, b_r, x) \longmapsto (b_1, a_2, \dots, a_r, b_r, x).$$

Neither operation is a complete strategy.

Computational evidence. The smallest useful witnesses are exact minimax states, not counterexamples to the linking theorem.

1. On $K_2 + d$, after

$$A : O_d, \qquad D : O_0, \qquad A : C_d,$$

   the defender faces $Q = (0)$ and $U = \{1\}$. The winning move is O_1; the apparently natural mirror close C_0 loses.

2. On the real path $3 - 0 - 1 - 2$ plus d, after

$$A : O_0, \qquad D : O_1, \qquad A : O_d,$$

   the defender faces $Q = (0, 1, d)$ and $U = \{2, 3\}$. Both available opens have the wrong live-score parity, and closing 0 loses. The unique winning move is the proactive wrong-parity open O_3, which creates an even odd-front corridor and stores a debt before the local trap appears.

Thus a state descriptor consisting only of current debt, front parity, parity classes in U, and one FIFO rotation is insufficient. The missing datum is a recursive repair certificate in the untouched graph. This is exactly what the affine flow is allowed to encode globally.

9 Local obstruction theory

The scalar cut potential gives a complete local classification, provided its limitation is stated explicitly. Regard the queue as a set when counting edges and put

$$P(Q, U) = e_G(Q, U) \pmod 2.$$

For a move m, let $\chi(m)$ be the change in P. If $R = Q \cup U$, direct edge accounting gives

$$\chi(O_x) = \deg_{G[R]}(x), \qquad \chi(C_f) = \deg_U(f), \qquad \chi(\text{pass}) = 0.$$

The middle term is the actual flip.

Proposition 24 (charge-matching and the poison state)**.** *Suppose $P = 0$, $|U|$ is even, and the position is nonterminal. After an opponent open, there is always a distinct legal open with the same χ. If a reply is due after an opponent close of charge c, there is a legal response of charge c except in exactly the following case:*

$$\begin{aligned} &c = 0, \qquad U \neq \varnothing, \qquad \deg_{R'}(u) = 1 \pmod 2 \quad \textit{for every } u \in U, \\ &Q' = \varnothing \quad \textit{or} \quad \deg_U(\operatorname{front} Q') = 1 \pmod 2. \end{aligned} \tag{41}$$

where $Q' = Q \setminus \{f\}$ and $R' = Q' \cup U$ after the close.

Proof. At $P = 0$,

$$\sum_{u \in U} \deg_R(u) = e(U, Q) = 0 \pmod 2,$$

because edges internal to U are counted twice. Hence the odd-charge opens form an even set; since $|U|$ is even, the even-charge opens do too. The charge class containing the opponent's opener therefore contains another vertex. Opening never has a ko obstruction, and the live set R is unchanged by an open, so this is a legal matching response.

If the opponent closes f with charge c, then

$$\sum_{u \in U} \deg_{R'}(u) = e(U, Q') = c \pmod 2.$$

For $c = 1$ an odd-charge open exists. For $c = 0$ an even-charge open exists unless every untouched vertex has odd charge; in that exceptional case a matching response still exists precisely when the new FIFO front exists and has even degree into U. A close clears ko, so that front close is legal. This is exactly the complement of (41). When U is empty all remaining closes, including the close after the unique terminal pass, have charge zero. □

Matching χ restores $P = 0$, but does *not* by itself pair the flip score: a mixed odd-close/odd-open round contains one actual flip. This is the debt which the global chain must transport. The dummy prevents (41) while it is untouched (it is an even-charge open) or is the new front (it is an even close), but it may be buried in the queue or already closed. The proposition therefore isolates the local obstruction without pretending to solve its recurrence.

One portion of that recurrence is nevertheless well founded. Let g be the accumulated flip debt and keep $P = e_G(Q, U)$ as above.

Lemma 25 (fixed-front poison descent)**.** *Suppose it is the defender's turn, $g = 1$, and the front f has positive even degree into U; ko may be set or clear. The defender can open a neighbour of f so that either the debt is discharged within the next attacker–defender round or the degree drops by exactly two. If the result is still positive the lemma may be repeated; if it is zero, the descended anticomplete front can be closed but need not discharge the debt. Hence the fixed-front loop terminates. Moreover*

$$H = g + P$$

is unchanged by every close. Thus $g = 1, H = 0$ excludes a singleton anticomplete front, although H is not generally preserved by opens.

Proof. Opening one neighbour changes $\deg_U(f)$ from even to odd and clears ko, since the queue was already nonempty. If the attacker closes f, that odd close discharges the debt. If the attacker opens a nonneighbour, the front remains odd and the defender closes it, again discharging the debt. The only remaining attacker move is to open a second neighbour; then the front is even again, the defender faces the same local problem when its degree remains positive, and exactly two of its untouched neighbours have been removed. At degree zero the neighbour-open loop stops; ko is clear at this descended boundary, so the even close is legal and carries no flip.

A close of charge c changes both g and P by c, so it fixes H. If $g = 1$ and $H = 0$, then $P = 1$, whereas a singleton queue whose front is anticomplete to U has $P = 0$. Finally an open x changes H by $\deg_{G[Q \cup U]}(x)$, which may be odd; therefore this exclusion is a local corridor condition, not a global invariant. □

The fixed-front data can be packaged much more richly and still fail to determine the continuation. For $Q = (q_0, \ldots, q_{m-1})$, define the *front signature*

$$\mathcal{F}(U, Q) = \Big(B|_{Q\times Q}, \Big\{\Big\{\big(B(u, q_0), \ldots, B(u, q_{m-1}); \deg_{G[U]}(u) \bmod 2\big) : u \in U\Big\}\Big\}\Big).$$

It retains the ordered queue graph, every live front degree, each untouched vertex's live-degree parity, and its full adjacency word into the queue; only the internal adjacency correlations among untouched vertices are forgotten, while their degree parities are retained.

Proposition 26 (front-signature insufficiency)**.** *The debt, mover, ko bit, and $\mathcal{F}(U, Q)$ do not determine zero-normalized safety. This already fails with five active vertices and a nonempty even queue.*

Proof. Put $Q = (0, 1)$, $U = \{2, 3, 4\}$, debt zero, ko clear, and let the attacker move. Compare

$$E(G_A) = \{23, 04, 14\}, \qquad E(G_B) = \{23, 24, 02, 12\}.$$

Neither graph has edge 01, and in both cases the untouched signature multiset is

$$\{\{(00; 1), (00; 1), (11; 0)\}\}.$$

For G_A the defender has the following first replies:

$$C_0 \mapsto C_1, \qquad O_2 \mapsto O_3, \qquad O_3 \mapsto O_2, \qquad O_4 \mapsto O_2.$$

The first branch leaves the edge 23 plus isolated vertex 4 on an empty queue; pair O_2/O_3, and after O_4 use O_2, opening the last vertex before any unmatched close. The middle two branches leave queue pairs $(0, 1)$ and $(2, 3)$ (in one order or the other): mate-closing gives equal charges, while an attack opening 4 makes every later close zero. In the last branch, after $Q = (0, 1, 4, 2), U = \{3\}$, answer C_0 by C_1 and O_3 by C_0; from $Q = (4, 2), U = \{3\}$ answer C_4 by O_3 and O_3 by C_4. Every displayed response has charge equal to the preceding attack, and the remaining closes have charge zero. Thus G_A is safe.

For G_B, the attack C_0 has charge one, forcing the sole normalized reply C_1, also of charge one. The queue is now empty. The attacker opens 2; ko forces the defender to open 3 or 4. In the first case C_2 has charge one into $\{4\}$, and in the second it has charge one into $\{3\}$; both legal defender replies then have charge zero. Hence G_B is unsafe despite the identical signature. $\square$

The first obstruction to same-action pairing does have a complete local repair. Write $B(u, v)$ for adjacency in the current graph.

Lemma 27 (two-switch tail)**.** *Suppose the defender moves just after an attacker close, the score debt is zero, and*

$$U = \{x, y\}, \qquad Q = (v_1, \ldots, v_{2r+1}).$$

If

$$B(v_{2i-1}, y) = B(v_{2i}, y) \quad (1 \le i \le r), \qquad B(v_{2r+1}, y) = B(x, y), \tag{42}$$

then the defender forces even final score with at most two changes of action type relative to the attacker's preceding move.

Proof. The defender first opens x, so the even queue is paired as $(v_1, v_2), \ldots, (v_{2r+1}, x)$ and every pair has equal adjacency to the sole untouched vertex y. While the attacker closes, the defender closes the next front; the two flip bits in each pair cancel by (42). If the attacker opens y, the defender closes the front. Now U is empty and every later close has charge zero. If the attacker never opens y before the pairs are exhausted, y is the harmless terminal singleton. The initial close/open and the possible later open/close are the two switches. $\square$

Computational evidence. Same-action adaptive pairing first fails, without a dummy, on two of the seven exceptional order-eight graphs: `GCZN^{` and `GEjt~{`. Call them G_1, G_2 in the table below. Exact minimax gives minimum worst-case switch count two on each. Their worst paths enter Lemma 27 at

	Q	U	(x, y)
G_1	$(1, 2, 4, 3, 5)$	$\{6, 7\}$	$(6, 7)$
G_2	$(1, 2, 3, 4, 5)$	$\{6, 7\}$	$(6, 7)$.

In both graphs 7 dominates the remaining live graph, so (42) is immediate. This proves the tail repair, not the still-missing prefix theorem that would force such a tail on every board.

10 The zero-normalized kernel and dynamic queue matchings

The order-eight search reveals a stronger scalar target than (36). Sample play after every defender move, with the odd-seeking player about to move and accumulated flip score zero. Away from the unique terminal-pass boundary, the queue then has even length. Put

$$\rho(U, Q) = |U| + \frac{|Q|}{2};$$

every two-move block decreases ρ by one.

Proposition 28 (zero-normalized repair recursion)**.** *Let Q have even length. There is a strategy returning the flip score to zero after every defender move if and only if the following recursive predicate* $\mathsf{Safe}(U, Q)$ *holds.*

- $\mathsf{Safe}(\varnothing, ())$ *holds.*
- $\mathsf{Safe}(\{z\}, ())$ *holds as the terminal macro O_z, forced zero-charge pass, C_z. The pass is the defender's move; after it the score is still zero, and the last close also has charge zero.*
- *After an attacker open O_z, a legal zero-charge reply is either O_w for some $w \in U - \{z\}$, leading to*
$$\mathsf{Safe}(U - \{z, w\}, Qzw),$$
or, when $Q = (f, Q_0)$ and $\deg_{U-\{z\}}(f) = 0$, the close C_f, leading to
$$\mathsf{Safe}(U - \{z\}, Q_0 z).$$
- *After an attacker close C_f of charge $\epsilon = \deg_U(f)$ from $Q = (f, h, Q_0)$, a same-charge reply is C_h when $\deg_U(h) = \epsilon$, leading to* $\mathsf{Safe}(U, Q_0)$*. When $\epsilon = 0$, the defender may instead open $w \in U$, leading to* $\mathsf{Safe}(U - \{w\}, hQ_0w)$*.*

Except inside the displayed terminal macro, every legal attacker move must have at least one listed reply whose child is safe.

Proof. All opens and passes have flip charge zero, and the only nonzero move charge is the displayed untouched degree of a FIFO front. Consequently the list contains exactly the replies whose charge equals the preceding attacker charge. The singleton macro is the only time the defender has no ordinary reply: opening the last untouched vertex onto an empty queue creates ko, so the forced pass clears it and leaves one final zero-charge close. Every other child has rank $\rho - 1$. Induction on ρ, treating the macro as a terminal leaf, proves sufficiency by selecting the recorded child after every attacker move. Conversely, record the first reply of any strategy which restores zero after each defender move, and apply the same argument to every reachable child. This produces the recursion. □

The resulting certificate is a ranked, branch-dependent repair forest. It is not yet a structural existence proof: defining the forest by the winning strategy would be circular. It does, however, state exactly what the finite search certifies and prevents weaker local invariants from being mistaken for the theorem.

The operational semantics has now also been made logically two-sided. Let $\mathsf{EvenWins}(s)$ be the existential/universal strategy tree above, and define $\mathsf{OddWins}(s)$ by exchanging the two seats and requiring terminal score one.

Proposition 29 (finite strategy duality)**.** *Every state admits exactly one of* $\mathsf{EvenWins}$ *and* $\mathsf{OddWins}$*. Consequently the linking theorem is pointwise equivalent to the nonexistence of an odd-forcing counterstrategy. This proposition, including the explicit dual trees, determinacy, incompatibility, and the equivalence, is formalized in* `formal/Ogdoad/Fifo.lean` *without axioms.*

Proof. The move rank strictly decreases. Induct on it. At a terminal state the score chooses exactly one tree. At a nonterminal state, if the designated seat moves, choose an even child when one exists; otherwise every child is odd by induction. If the other seat moves, exchange "even" and "odd". The two trees cannot coexist: at a choice node follow the chosen child through the opposite tree's universal node, and continue by induction. The terminal scores then contradict one another. □

There is an exact terminal corridor inside the repair forest. Suppose $U = \{b\}$, the attacker moves, and the even queue is $Q = (v_0, v_1, \ldots, v_{2r-1})$. Put $c_i = B(v_i, b)$. Scan consecutive cells $c_{2i}c_{2i+1}$ from the front. The state is zero-normalized safe exactly when the first cell which is not 11, if one exists, is not 10. Indeed, an attacker OPEN of b empties U. Against C_{v_0}, a zero first bit is answered by O_b; a one first bit has the unique normalized form C_{v_1}, which succeeds exactly when the second bit is also one and then recurses on the tail. This is a two-step induction on Q. The sufficient strategy, together with the all-touched terminal tail, is formalized as `SingletonTail` and `evenWins_singletonTail` in the same Lean file.

This exact tail does *not* promote a bounded-defect matching invariant to the full game.

Proposition 30 (unbounded absorbable defect fan)**.** *For every $r \geq 1$ there is a normalized reachable checkpoint whose queue consists of r balanced cells, but after the attacker opens the dummy, every normalized reply makes all r old cells defective at once. The branch is nevertheless zero-normalized safe in one more response square.*

Proof. Take

$$Q = (x_1, y_1, \ldots, x_r, y_r), \qquad U = \{a, b, d\},$$

where d is isolated, and put in exactly the edges $x_i a$ and $y_i b$. Opening the displayed queue in order reaches the checkpoint, and every cell has charge 11 into U. After the attacker plays O_d, closing x_1 has charge one, so a normalized defender must play O_a or O_b. After O_a the residual set is $\{b\}$ and all old cells read 01; after O_b they all read 10. Thus the defect count is r, with either orientation.

Choose O_a. The attacker can now only open b or close x_1, whose charge has become zero. Answer O_b by C_{x_1} and answer C_{x_1} by O_b. Either way U becomes empty with score zero, so every later close is neutral. This is precisely the first-cell-01 exit of the singleton corridor. $\square$

Therefore clean consecutive matching, a single transported defect, monotone defect position, and every uniformly bounded defect count are false induction objects. What must be transported is a *factored* repair fan: arbitrarily many local defects can share one complementary OPEN/CLOSE absorber. The open factor-extension theorem must recognize such shared absorbers while still surviving attacker pruning.

Computational evidence. Every even-order graph through order eight has $\mathsf{Safe}(V, ())$. On all 12,346 order-eight isomorphism classes, the second player can therefore keep the *actual* flip score zero after every one of her moves. The first failures of the stronger same-action rule occur on `GCZN^{` and `GEjt~{`; the zero-normalized strategy survives by using a close/open or open/close rotation. Static selectors fail more sharply: on `G?bFf[` an opener's unique same-(p, b) mate is unsafe, and on `GCz^vw` an odd-degree opener's only safe reply has even degree. These are failures of proposed selectors, not of Proposition 28 or the linking theorem.

The normalized repair forest cannot itself be promoted to the Euler quotient target of Theorem 16. *Computational evidence.* On eight vertices, let the primary and secondary Euler graphs have edge sets

$$A = \{02, 03, 06, 07, 13, 15, 23, 24, 26, 35, 36, 37, 47, 56, 57\},$$
$$B = \{01, 04, 06, 07, 12, 13, 15, 16, 17, 24, 25, 27, 35, 45, 47, 56, 57, 67\}.$$

Both degree vectors vanish modulo two, and A has a zero-normalized repair forest. Nevertheless, the exact deterministic attack generated by `experiments/linking_game.py::normalized_secondary_moment` has 88 complete A-normalized response histories and every one satisfies

$$\langle B, D(h)\rangle = 1.$$

Therefore the affine hull of that normalized response set misses $\mathrm{Cut}(V)$ by Theorem 16. This does not refute scalar zero normalization for A; it refutes the stronger move of first pruning by one graph's zero-score replies and then asking the surviving carrier for a graph-independent cut-valued contraction. A proof must use the scalar target without treating normalization as a label-independent chain map.

The rotations have an exact ordered-matching interpretation. At a checkpoint write

$$Q = (v_0, v_1, \ldots, v_{2r-1})$$

and match consecutive queue vertices. Pair opens extend this matching and pair closes delete its front edge. If the attacker appends $z = v_{2r}$ and the defender gives the legal zero close C_{v_0}, the new matching is shifted by one place. Let

$$P(W) = \sum_{i=0}^{2r-1} [v_i, v_{i+1}], \qquad W = (v_0, \ldots, v_{2r}).$$

Then the old and shifted matching chains differ by $P(W)$ and

$$\partial P(W) = e_{v_0} + e_z.$$

More precisely, with the triangle fan

$$F(W) = \sum_{i=1}^{2r-1} [v_0, v_i, v_{i+1}],$$

one has

$$P(W) + [v_0, z] = \partial F(W). \tag{43}$$

For the adjacency 1-cochain B, therefore,

$$\langle B, P(W)\rangle = B(v_0, z) + \sum_{i=1}^{2r-1} \delta B(v_0, v_i, v_{i+1}). \tag{44}$$

The second term is genuine two-graph holonomy. Endpoint exchange, a queue matching span, or a symplectic basis invariant sees only $B(v_0, z)$ and is therefore insufficient on an arbitrary graph.

The triangle shears are coherent but not flat. If $\tau(i, j, k) = \delta B(i, j, k)$, then

$$\tau(j, k, l) + \tau(i, k, l) + \tau(i, j, l) + \tau(i, j, k) = 0, \tag{45}$$

which is $\delta^2 B = 0$. Thus the ambient complete simplex fills the fan in (43) consistently. The adversarial obstruction is in its coefficients: after fixing an attacker strategy, compatible histories are not closed under deleting an interior letter. Defender coefficients may split, but an attacker coefficient must follow one prescribed child. Projection to compatible histories consequently fails to commute with the simplicial boundary.

There is a stronger obstruction before attacker pruning. On the score-forgotten public-state graph give a real close C_f from untouched set U the universal charge

$$c(U, f) = \sum_{u \in U \cap R} fu \in \mathcal{E}_R,$$

and give every other move charge zero. On $M = \mathbb{F}_2 \oplus \mathcal{E}_R$ transport along a move of charge c by $T_c(a, z) = (a, z + ac)$.

Proposition 31 (universal public-state holonomy)**.** *Let $Q = (f, Q_0)$ be ko-clear, let $Q_0 \neq ()$, and let $z \in U \cap R$. The legal paths*

$$O_z; C_f \qquad \text{and} \qquad C_f; O_z$$

end in the same score-forgotten public state, but their transports differ by T_{fz}. Every real basis edge fz occurs on a reachable such square. Consequently, if $p : \mathcal{E}_R \to W$ is a linear quotient on which the induced transports are flat on every schedule square, then $p = 0$. In particular scalarization by a graph B is flat on all squares if and only if B is edgeless.

Proof. The close-before-open path has charge $c(U, f)$, whereas the open-before-close path has charge $c(U - \{z\}, f)$. Their difference is exactly fz. The identity $T_c T_{c'} = T_{c+c'}$ proves the transport assertion, and the two paths visibly leave the same untouched set, queue, ko bit and mover.

For distinct real f, z, first play O_f, O_d, where d is the isolated dummy. The resulting queue is (f, d), ko is clear, and z is untouched, so the displayed square is reachable and has curvature fz. These fz form a basis of $\mathcal{E}_R$. Flatness after applying p therefore gives $p(fz) = 0$ basiswise and hence $p = 0$. Pairing with a scalar adjacency vector B sends this curvature to $B(f, z)$, which vanishes for all pairs exactly when B has no edges. $\square$

The qualification $Q_0 \neq ()$ is exact. At a singleton queue the two paths have different ko bits, so there is no square to fill. Squares involving the dummy have zero real curvature, and the missing singleton square is precisely where the dummy cannot cone off a nonzero fz. Hence adding ordinary curvature fillers to the public-state complex either fails to define a flat complex or, after quotienting, erases the entire terminal invariant. The Lean model kernel-checks both the charge identity and the complete operational square, including reachability of every real basis coordinate, in `Ogdoad/Fifo.lean`.

Eulerian support makes this curvature into an honest Euler complex. Recall that a 3-oik is a multiset of three-element rooms in which every two-element wall occurs with even multiplicity [8].

Theorem 32 (Euler curvature oik)**.** *Let G be an Eulerian graph of even order n, with adjacency* 1*-cochain B, and put*

$$\tau(i,j,k) = B(i,j) + B(i,k) + B(j,k).$$

The triples on which $\tau = 1$ form a 3*-oik. Moreover this binary room multiset has a coherent oriented integral multiset lift: there is an integral* 2*-cycle z in the full simplex whose coefficient on each triangle is congruent to τ modulo* 2*.*

Proof. For a wall $\{i,j\}$, its room incidence modulo two is

$$\sum_{k \neq i,j} \tau(i,j,k) = (n-2)B(i,j) + (\deg i - B(i,j)) + (\deg j - B(i,j)) = 0,$$

because n, $\deg i$, and $\deg j$ are even. This is exactly the oik wall condition.

For the oriented lift, choose an orientation on every triangle and regard the $0/1$ room set as an integral 2-chain t. The wall condition says that ∂t is even, so write $\partial t = 2b$. Then $\partial b = 0$. The full simplex has $H_1(-;\mathbb{Z}) = 0$, hence $b = \partial c$ for an integral 2-chain c. The chain $z = t - 2c$ is an integral cycle and satisfies $z \equiv t \equiv \tau \pmod 2$. Replacing each nonzero coefficient by that many signed copies gives the claimed coherent oriented multiset. ☐

Equation (44) now says exactly that the phase defect is the parity of rooms of this 3-oik occurring in the triangle fan $F(W)$. This is a genuine bridge to complementary pivoting, but ordinary oik parity does not close the game. A fan overlaps in vertices and is not a room partition; attacker pruning is irreversible and need not give every wall even incidence; and a pivot path pairs endpoints, producing augmentation zero over $\mathbb{F}_2$, whereas the required response chain has augmentation one. The curvature also cannot be oriented on Markov states alone. The two legal words $C_f O_z$ and $O_z C_f$ end in the same state when both orders are legal, but their charges differ by $B(f,z)$. Thus a successful use of Theorem 32 needs a relative, strategy-pruned, charge-decorated oik contraction, not the constant-coefficient room-partition theorem.

The full simplex nevertheless gives an exact dichotomy. Write J for the all-edge cochain and $\epsilon(R) = \langle J, R\rangle$ for the number of edges in a binary chain modulo two.

Theorem 33 (odd isotropic cycle dichotomy)**.** *Let G be an Eulerian graph of positive even order, with adjacency chain B. Exactly one of the following holds.*

1. *There are distinct vertices i,j,k spanning an even number of edges of G, or equivalently*

$$\tau(i,j,k) = B(i,j) + B(i,k) + B(j,k) = 0.$$

Thus the single triangle $F = [i,j,k]$ and its boundary $R = \partial F$ obey

$$\epsilon(F) = \epsilon(R) = 1, \qquad \langle \delta B, F\rangle = \langle B, R\rangle = 0.$$

2. *There is an odd set $S \subseteq V$ such that*

$$B = J + \delta 1_S, \qquad G = K_S \sqcup K_{V \setminus S}.$$

Both cliques have odd order. This exceptional family satisfies the parity-cell hypothesis of Theorem 17 for every partition into pairs, and hence every prescribed ordered-pair checkpoint is safe.

Proof. Assume first that $|V| \geq 4$. The coboundary δJ is one on every triangle. If the first alternative fails, then $\delta B = 1 = \delta J$ on every triangle, so $\delta(B + J) = 0$. The 1-cocycles of the full simplex are exact, hence $B + J = \delta 1_S$ for some $S \subseteq V$. Conversely, that identity gives $\delta B = \delta J = 1$ on every triangle, so the alternatives are exclusive. When the first alternative holds, the displayed claims for F and R are the augmentation identity for one triangle and Stokes.

Put $s_v = 1_S(v)$. Because $|V|$ is even, at every vertex

$$\deg_B(v) = \sum_{u \neq v} (1 + s_u + s_v) = 1 + |S| \pmod 2.$$

Eulerianity therefore makes $|S|$ odd. Moreover $B(u, v) = 1 + s_u + s_v$ is one exactly when u, v lie on the same side of the cut, proving the two-clique description; its complement also has odd order. At order two, Eulerianity forces the empty graph, which has the same form for either singleton S.

Finally, in the exceptional family, if C, D are any two-element cells, then

$$e_G(C, D) = |C \cap S||D \cap S| + |C \cap S^c||D \cap S^c| = 0 \pmod 2,$$

because a two-element cell has equal intersection parities with S and S^c. Any perfect matching containing a prescribed pair therefore invokes Theorem 17 from that ordered-pair checkpoint. $\square$

Outside the solved two-clique family, Theorem 33 supplies exactly the odd, zero-curvature full-simplex chain that ordinary oik parity was missing—in fact, one even-curvature triangle suffices. It still does not solve the game: that face need not be compatible with a fixed attacker strategy. The remaining theorem is now sharply one of promoting such a triangle through attacker pruning while preserving odd augmentation and its cut-valued terminal moment.

The wall parity itself has a precise game-facing form. It supplies an odd fan of locally cancelling OPEN replies, but the continuation coefficients are still the unresolved datum.

Lemma 34 (same-curvature odd wall fan)**.** *Let G be an even-order Eulerian graph, fix distinct x, y, and put*

$$\tau_{xy}(v) = B(x, y) + B(x, v) + B(y, v) \qquad (v \notin \{x, y\}).$$

Each of the two τ_{xy} fibres in $V - \{x, y\}$ has even cardinality. Consequently, after an attacker opens z from the ordered-pair checkpoint $Q = (x, y)$, the set

$$W_z = \{w \notin \{x, y, z\} : \tau_{xy}(w) = \tau_{xy}(z)\}$$

has odd cardinality. For every $w \in W_z$, if the defender opens w and the attacker next closes x, then closing y gives the same charge and cancels that close pair.

Proof. Because $|V|$, $\deg x$, and $\deg y$ are even,

$$\sum_{v \neq x, y} \tau_{xy}(v) = (|V| - 2)B(x, y) + (\deg x - B(x, y)) + (\deg y - B(x, y)) = 0.$$

Thus the one-fibre is even; its complement in the even set $V - \{x, y\}$ is even as well. Removing z from its own fibre leaves the odd set W_z.

After z, w have opened, put $U = V - \{x, y, z, w\}$. Eulerianity gives

$$\begin{aligned}\deg_U(x) + \deg_U(y) &= B(x, z) + B(y, z) + B(x, w) + B(y, w) \\ &= \tau_{xy}(z) + \tau_{xy}(w) = 0.\end{aligned}$$

These are exactly the charges of the consecutive closes of x and y, so they agree and cancel. $\square$

Lemma 34 is the local 3-oik wall promised by Theorem 32; it is not a contraction of the response tree. The order-eight open-completion witness below has an attacker open for which every OPEN child can be made losing while the unique safe reply is the phase-changing close. A valid odd flow must therefore choose a strategy-dependent subset across OPEN and CLOSE children and across earlier front levels, rather than assigning coefficient one to the whole odd wall.

The wall does, however, transport an odd flow through an arbitrarily long run of OPEN moves while one ordered front pair is fixed. The right hypothesis is slightly more general than Eulerianity. For a checkpoint $Q = (x, y, Q_0)$, put

$$\eta_{xy}(u) = B(x, u) + B(y, u) \qquad (u \in U),$$

and call the front pair *balanced* when both η_{xy} fibres in U have even cardinality.

Theorem 35 (balanced-front odd-flow transport)**.** *Fix a balanced checkpoint with the attacker to move and fix an arbitrary deterministic attacker strategy until x closes. There is a canonical response flow of odd augmentation with the following rules:*

- *after an attacker OPEN O_z, put coefficient one on every reply O_w with $\eta_{xy}(w) = \eta_{xy}(z)$;*
- *after the attacker closes x, reply by closing y and exit the fixed-front phase.*

Every exit has equal charges on C_x, C_y, so the phase contributes zero scalar score. This assertion and the flow construction also allow one of x, y to be the isolated dummy, with every dummy edge projected to zero. If U_λ is the untouched set at an exit λ, its future disjointness increment on the deleted pair is exactly

$$\kappa_{xy}(U_\lambda) = \sum_{u \in U_\lambda} (xu + yu). \tag{46}$$

When both front vertices are real this chain is Eulerian; in every case its real projection is B-isotropic:

$$x, y \in R \Longrightarrow \partial \kappa_{xy}(U_\lambda) = 0, \qquad \langle B, \kappa_{xy}(U_\lambda) \rangle = 0.$$

Proof. If the attacker opens z, its η-fibre had even size, so after removing z the displayed reply set is odd and nonempty. Every selected child removes two vertices of one fibre; hence both fibres remain even and the construction may be repeated. Each OPEN round removes two untouched vertices, so the process terminates. Attacker nodes route their coefficient to the one child prescribed by the strategy, while every defender fan has odd augmentation; the exit flow therefore has augmentation one.

At an exit,

$$\deg_{U_\lambda}(x) + \deg_{U_\lambda}(y) = \sum_{u \in U_\lambda} \eta_{xy}(u) = 0,$$

so the consecutive close charges agree. A residual untouched vertex is disjoint from both deleted front intervals, whereas every vertex already in the queue overlaps them. This proves (46). If x, y

are real, the residual set has even cardinality, so every vertex in the displayed complete bipartite chain has even degree. Pairing the real projection with B gives the same even η-sum whether or not one front vertex is the dummy, proving the last two assertions with the stated qualification. $\square$

At an Euler ordered-pair checkpoint, balancedness follows from Lemma 34, since $\eta_{xy} = \tau_{xy} + B(x, y)$. Theorem 35 is therefore a genuine strategy-pruned local coefficient transport, not merely the one-round wall count. Its nonzero κ is the local coefficient which must be carried to the next FIFO front.

That coefficient has an exact cell-cut calculus. For arbitrary disjoint sets P, X in a residual live graph H, write

$$K(P, X) = \sum_{p \in P,\, x \in X} px.$$

It is an Eulerian edge chain when both sets are even. The application below uses two-cells $P_i = \{a_i, b_i\}$ whose endpoints have equal degree parity in H.

Proposition 36 (cell-cut transport)**.** *Let the residual queue be $P_1 P_2 \cdots P_r$, where $P_i = \{a_i, b_i\}$ and $\deg_H(a_i) = \deg_H(b_i) \pmod 2$ for every i, and let U be untouched. For every prefix $A_k = P_1 \cup \cdots \cup P_k$,*

$$\sum_{i \le k} K(P_i, U) + \sum_{i \le k < j} K(P_i, P_j) = \delta_H(A_k). \tag{47}$$

Consequently, with

$$\Delta_i = \langle B, K(P_i, U) \rangle, \qquad c_{ij} = \langle B, K(P_i, P_j) \rangle,$$

one has

$$\sum_{i \le k} \Delta_i = \sum_{i \le k < j} c_{ij}. \tag{48}$$

In particular the number of cells with $\Delta_i = 1$ is even. Moving a new two-cell R from U to the back of the queue obeys

$$K(P, U - R) = K(P, U) + K(P, R). \tag{49}$$

Thus OPEN converts untouched κ-mass into a queued four-cycle, while successive CLOSE pairs, with intervening OPEN transfers accounted for by (49)*, move the same defect across the phase cuts in* (47)*; after the entire queue closes only a cut remains.*

Proof. Both sides of (47) are exactly the complete bipartite edge chain from A_k to $H - A_k$: the first sum contains its edges to U and the second its edges to later queue cells. Pairing with B proves (48), because

$$\langle B, \delta_H(A_k) \rangle = \sum_{i \le k} \big(\deg_H(a_i) + \deg_H(b_i)\big) = 0.$$

More explicitly, $\deg_H(a_i) + \deg_H(b_i) = 0$ gives

$$\Delta_i = \sum_{j \ne i} c_{ij};$$

summing over i cancels every c_{ij} twice and proves the even-defect claim. Equation (49) is the disjoint union $P \times U = (P \times (U - R)) \sqcup (P \times R)$ over $\mathbb{F}_2$. $\square$

For the Euler ordered-pair exits of Theorem 35, the homogeneity hypothesis is automatic. Put $H = G - \{x, y\}$. Every wall-created cell $P_i = \{z_i, w_i\}$ has $\eta_{xy}(z_i) = \eta_{xy}(w_i)$, while Eulerianity gives

$$\deg_H(v) = B(v, x) + B(v, y) = \eta_{xy}(v).$$

Moreover U is even, so each $K(P_i, U)$ is itself Eulerian. This application does not extend merely from balancedness at an arbitrary non-Euler checkpoint; the cell-degree hypothesis is an additional datum there.

The tempting next induction—that the first residual cell is balanced after summing the canonical wall exits—is false. *Computational evidence.* On the Euler graph

$$E = \{02, 03, 04, 07, 24, 34, 36, 37, 45, 46, 47, 57\},$$

start at $(x, y) = (0, 1)$. The nonzero η_{01} fibre is $\{2, 3, 4, 7\}$ and the zero fibre is $\{5, 6\}$. Let the attacker open 2; after replies $3, 4, 7$, respectively, let it follow

$$C_0, \qquad O_6O_7C_0, \qquad O_5C_0.$$

The canonical transport exits at

	Q	U
(a)	$(2, 3)$	$\{4, 5, 6, 7\}$
(b)	$(2, 4, 6, 5, 7, 3)$	$\varnothing$
(c)	$(2, 7, 5, 6)$	$\{3, 4\}$.

Their first-cell imbalances are $0, 0, 1$. In the last exit both cells 27 and 56 are imbalanced and $c_{27,56} = 1$: the defect is not lost, but sits on the intermediate phase cut exactly as (48) predicts. Hence the remaining lemma must causally lift these cross-cell four-cycles through legal OPEN/CLOSE matching shifts; aggregate front balance is not the invariant.

Even the complete ordered cell-cut datum is not a Markov state for this lifting. On seven real vertices, with isolated $d = 7$, put

$$\begin{aligned} E_{\text{bad}} &= \{01, 02, 04, 12, 13, 14, 16, 23, 24, 26, 34, 35, 45, 56\}, \\ Q &= (6, 1, 5, 4), \qquad U = \{0, 2, 3, 7\}. \end{aligned}$$

All four individual queue charges are one, all four queued vertices have odd full degree, both cell defects Δ_i vanish, and $c_{61,54} = 0$. Nevertheless the checkpoint is unsafe even for the unrestricted terminal-score game. The attacker opens the isolated dummy. Replies O_0 and O_3 are met by the odd close C_6, while C_1 and every remaining OPEN have charge zero. After reply O_2, the attacker closes 6 evenly; the three normalized replies O_0, O_3, C_1 are defeated respectively by short continuations beginning C_1, C_1, C_5. In the O_0 subbranch the forced charge-one responses continue through C_5, C_4. Direct evaluation gives one on each displayed spoiler and zero on every available response; the exact tree is replayed by the checker.

This is not an arbitrary unreachable carrier. Add old fronts $x = 8, y = 9$, make x isolated, and join y to $\{0, 1, 2, 4, 5, 6\}$, the odd-degree set of E_{bad}. The resulting ten-vertex graph is Eulerian. From $Q = (x, y)$ the canonical balanced-wall branch

$$O_6, O_1, O_5, O_4, C_x, C_y$$

exits exactly at the displayed bad checkpoint with zero accumulated charge. Thus no proof may continue every canonical wall exit independently.

For comparison, the labelled graph

$$E_{\text{good}} = \{04, 06, 13, 14, 15, 23, 35, 45\}$$

has the same Q, U, the same four individual charges, full-degree colours, Δ vector, cross-cell parity, and formal chains $K(P_i, U), K(P_1, P_2)$, but exact zero-normalized recursion marks it safe. *Computational evidence.* The graph6 records are respectively `Fz{ZG` and `FBp[?`; `experiments/linking_game.py` verifies both decisions and the Euler lift. Hence vertex-resolved continuation data below the cell aggregation is indispensable.

Nor is the Eulerian case a simplifying restriction of the scalar kernel.

Theorem 37 (Euler ordered-pair kernel is complete). *Suppose that for every even-order Eulerian graph G and every ordered pair of distinct vertices (x, y), the checkpoint*

$$U = V(G) - \{x, y\}, \qquad Q = (x, y)$$

is safe in the zero-normalized recursion. Then every even-order graph H satisfies $\mathsf{Safe}(V(H), ())$. Equivalently, any even-board counterexample produces an Eulerian ordered-pair counterexample two vertices larger.

Proof. Let H have nonempty even vertex set V, put $p(v) = \deg_H(v) \pmod 2$, and let $O = \{v : p(v) = 1\}$. Handshaking makes $|O|$ even. Choose any odd set $A \subseteq V$ and put $C = A \triangle O$. Adjoin x, y, the edge xy, all edges from x to A, and all edges from y to C. For $v \in V$,

$$\deg_G(v) = p(v) + 1_A(v) + 1_C(v) = 0 \pmod 2,$$

while $\deg_G(x) = 1 + |A|$ and $\deg_G(y) = 1 + |C| = 1 + |A| + |O|$ are even. Thus G is Eulerian of even order.

At the displayed checkpoint, the attacker may close x with charge $|A| = 1$. Zero normalization forbids an open reply, and FIFO leaves only the close of y, whose charge $|C|$ is also one. The two charges cancel and the resulting state is exactly $U = V$, $Q = ()$, attacker to move, with score zero in the original graph H. Safety of the Eulerian checkpoint therefore implies $\mathsf{Safe}(V, ())$; contraposition gives the final assertion. $\square$

This is not repaired by ordinary topology of injective words. Those words form a Boolean cell complex with

$$\partial[v_0 \cdots v_k] = \sum_i [v_0 \cdots \widehat{v}_i \cdots v_k] \quad \text{over } \mathbb{F}_2$$

[6, 7]. Define the strategy carrier here as the Boolean subcomplex generated by the attacker-compatible pre-close tail words. Even with a dummy, attacker pruning need not leave this carrier acyclic. On $K_2 + d$, let the attacker open a real front f and, after the forced-open reply is chosen as O_a or O_d, close f immediately. Before the first close the carrier consists of the two isolated vertices $[a]$ and $[d]$; the joining cells $[ad]$ and $[da]$ have been pruned. Its reduced H_0 is nonzero. The linking response still exists because the two close charges differ. Hence any successful carrier theorem must retain the charge and continuation labels; ordinary shellability or a constant-coefficient cone is not enough.

Three further exact obstructions delimit the remaining Boolean contraction. First, the two-bit colour is a prefix moment, not a winning selector. *Computational evidence.* On the eight-vertex graph

$$E = \{01, 03, 05, 07, 12, 14, 15, 17, 24, 26, 27, 34,$$
$$37, 45, 46, 47, 57, 67\},$$

vertices $6, 7$ have the same (p, b) colour, but Q_{67} is unsafe while Q_{76} is safe. The unique first spoiler in Q_{67} is O_5. Vertices $6, 7$ have odd degree while 5 has even degree, so the local unsafe-Q hypotheses used by odd routing do not themselves force the witness choice $y \mapsto z_y$ to preserve Y. Functional-digraph cycle parity would first need a genuinely least-counterexample-specific closure theorem. There is also an opener with no safe same-colour mate: in graph6 class `G?ben[`, with

$$E = \{04, 05, 15, 25, 06, 16, 36, 56, 07, 17, 27, 47, 57, 67\},$$

vertex 2 has sole same-colour mate 7, yet Q_{27} is unsafe. Its unique first spoiler is O_3; the reverse Q_{72} is safe. Thus neither orientation nor existential choice within one colour class proves the root theorem.

Second, a safe residual strategy cannot always be lifted by insisting on an open response. On

$$\begin{aligned} E = \{&01, 02, 03, 04, 05, 06, 12, 13, 14, 17, 23, 26, \\ &27, 35, 37, 56, 57\}, \end{aligned}$$

the checkpoint

$$Q = (4, 3), \qquad U = \{0, 1, 2, 5, 6, 7\}$$

is safe in the zero-normalized recursion. After attacker O_6, the unique safe reply is the zero close C_4, leading to $Q = (3, 6)$ and $U = \{0, 1, 2, 5, 7\}$. The five open replies by $0, 1, 2, 5, 7$ are all unsafe; respective short spoilers may be chosen as C_4, C_4, O_7, O_7, O_1. Hence the phase-changing OC branch is not an artifact of the finite search: it is indispensable even when $|U|$ was even.

Finally, time reversal does not repair the strategic pruning. For a pass-free complete event word put

$$\rho(e_1 \cdots e_m) = \bar{e}_m \cdots \bar{e}_1, \qquad \overline{O_v} = C_v, \quad \overline{C_v} = O_v.$$

This is a legal FIFO schedule, swaps the two seats, and preserves D coordinatewise. Indeed, if both opening and closing order in the original word are $v_1, \ldots, v_n$, both orders after reversal are $v_n, \ldots, v_1$, so FIFO is preserved. A pass-free word has no adjacent singleton block $O_v C_v$, hence reversal creates no ko violation. Position i moves to $2n + 1 - i$, which swaps seat parity, while each activity interval $[o_v, c_v]$ reflects to $[2n + 1 - c_v, 2n + 1 - o_v]$; interval disjointness, and therefore D, is unchanged.

Let $\mathcal{L}_{\mathrm{pf}}$ be the full set of legal pass-free complete schedules and extend ρ linearly to $\mathbb{F}_2[\mathcal{L}_{\mathrm{pf}}]$. For at least two vertices, ρ is fixed-point-free: a fixed word would have its common FIFO opening/closing order equal to its reverse, impossible for distinct labels. Thus every ρ-invariant chain in this full space is a sum of two-element orbits, hence lies in the image of $1 + \rho$. For every chain c,

$$(\mathrm{aug}, D)((1 + \rho)c) = (0, 0).$$

Reversal averaging therefore produces a homogeneous zero-moment chain, never the required affine certificate $(1, 0)$. Even a hypothetical causal reversal would preserve the odd payoff rather than complement it, so composing two odd-target strategies would give no contradiction.

This identity does not extend to a strategy contraction. If one attempts to set $\rho(P) = P$, a terminal singleton component O_x, P, C_x moves to the beginning; when other vertices remain untouched there, its pass is not forced and is illegal. Even without passes, fixed strategies are anti-causal under ρ. Put the original attacker in the even positions. One deterministic attacker strategy contains both histories below: after O_0 it prescribes O_d; after the defender's choice C_0 or O_1 it prescribes respectively O_1 or C_0; and after C_d it prescribes C_1. The smallest such local witness has real vertices $0, 1$ plus d:

$$\begin{array}{r|l} h_C = O_0 O_d C_0 O_1 C_d C_1 & D(h_C) = 01, \\ h_O = O_0 O_d O_1 C_0 C_d C_1 & D(h_O) = 0. \end{array}$$

The two histories reconverge across the commuting C_0/O_1 diamond, whose label difference is the edge coordinate 01. Reversal gives a common prefix O_1O_d but demands different moves, C_1 and O_0, at the same odd-position attacker node. Hence the reversed pair cannot lie under one deterministic reversed attacker strategy. Separately, the diamond involution ι exchanging C_0 and O_1 swaps h_C and h_O, so

$$\mathbb{F}_2\{h_C, h_O\}^\iota = \{0, h_C + h_O\}.$$

Both invariant chains have even augmentation (and $D(h_C + h_O) = 01$), whereas the valid odd zero-moment certificate is the nonsymmetric singleton h_O. Diamond symmetrization therefore destroys the odd augmentation rather than proving it.

The dummy does have one exact phase interpretation when it is opened onto an empty root.

Proposition 38 (dummy-first draft and switch)**.** *Suppose $R \neq \varnothing$ and the first move is O_d. After the ko-forced first real OPEN, the play until C_d is precisely an alternating draft of distinct real vertices: on each turn the player either OPENs an untouched real vertex or plays the zero-charge switch C_d. At the switch, ordinary no-dummy FIFO play begins at score zero from the nonempty drafted queue, with ko clear and with the other player to move. If all real vertices are drafted before the switch, C_d is forced. Conversely every such draft-and-switch word is legal.*

Proof. Opening d onto the empty queue sets ko, so the next move is a real OPEN. Until d closes it remains the FIFO front. Hence the only legal CLOSE is C_d, while every other legal move is an OPEN of a fresh real vertex. All these OPENs and C_d have charge zero. Closing d clears ko and leaves, in its opening order, the nonempty queue of drafted real vertices. These observations also prove the converse and the exhausted-draft clause. □

Thus the dummy is a genuine one-use phase switch at a dummy-first root, but not an empty-root reduction. In particular, the stopped close-first theorem cannot be invoked at the switch: its hypothesis is a defender-first empty no-dummy root, whereas Proposition 38 exposes a nonempty queue and either player may have taken the switch. The nonempty drafted prefix, rather than the dummy event itself, remains part of the global ancestry problem.

For a general history, deleting the dummy has one further, exactly localized defect.

Proposition 39 (dummy deletion and the unique ko wall)**.** *Let h be a complete legal history with isolated dummy d. Form the raw projected event word by deleting O_d, C_d and any pass, while retaining the real events in order. Then:*

1. *every real CLOSE has the same charge before and after deletion;*
2. *relative to the raw projected positions, the controller parity of precisely the real events strictly between O_d and C_d is reversed;*
3. *after inserting the unique forced pass exactly when required at a terminal singleton, the projected word is a legal real FIFO history except possibly at one internal ko wall; and*
4. *if that wall occurs, its consecutive real events are O_v, C_v, O_w. Replacing them by O_v, O_w, C_v gives a legal real history $\widetilde{h}$ and*

$$D_R(\widetilde{h}) = D_R(h) + vw, \qquad F_G(\widetilde{h}) = F_G(h) + B(v, w).$$

The wall occurs exactly when d is the sole open vertex immediately before O_v, and both C_d and C_v occur before the next real OPEN O_w.

Proof. Because d is isolated, removing it from the untouched set changes no real CLOSE charge. Deleting two event positions preserves controller parity outside their open interval and reverses it inside.

Deletion preserves the real opening order, closing order, and FIFO condition. Suppose nevertheless that a projected O_v opens an empty real queue and is followed by C_v while another real vertex remains untouched. The original O_v could not have opened an actually empty queue, since its ko would have made C_v illegal. Thus d was the unique open vertex before O_v. FIFO then forces C_d before C_v, and no other real event can intervene before the next real OPEN O_w. Once this happens d is closed, so there can be no second internal wall. If no such w exists, the projected block is terminal and inserting its forced pass is the unique repair.

In the internal case, moving O_w immediately before C_v clears ko and preserves FIFO legality. The adjacent exchange changes no real interval relation except that of v, w: their intervals were disjoint and now overlap. This proves both displayed identities and the characterization of the wall. $\square$

Consequently the dummy cannot simply be erased to make a shadow copycat game. If π deletes O_d, C_d, then $\pi(h_C) = O_0C_0O_1C_1$ is illegal: after O_0, ko forbids C_0 while O_1 remains available. Even when a schedule and its dummy deletion are both legal, controllers are not preserved. For

$$k = O_dO_0O_1C_dC_0C_1, \qquad \pi(k) = O_0O_1C_0C_1,$$

the projected real events were controlled in order by B, A, A, B, not by alternating seats. With only one real vertex, deletion instead requires the state-dependent terminal pass from Proposition 39. Hence dummy deletion preserves neither legality, turn ownership, nor the pass boundary; it is not a game-tree morphism.

The failure is concentrated at an exact ko wall. Suppose d is untouched, $Q = (f, Q_0)$ has real front f, and ko is clear. The adjacent words

$$O_dC_f, \qquad C_fO_d$$

have the same real score and end with the same untouched set and queue Q_0d. Their ko bits agree unless Q_0 is empty; at that singleton wall the first word ends with ko clear and the second with ko set. Dually, from $Q = (d, Q_0)$ and untouched x, the words C_dO_x and O_xC_d end in the same queue Q_0x with equal real score, but at $Q_0 = ()$ their ko bits are set and clear respectively. This follows directly because a close clears ko, whereas an open sets it exactly when the pre-open queue is empty.

The first wall is already label-essential. The legal word

$$O_0O_dC_0O_1C_dC_1$$

has $D_R = 01$, whereas deleting the dummy gives the illegal word $O_0C_0O_1C_1$. The only legal real FIFO word with the same real opening and closing orders is $O_0O_1C_0C_1$, whose disjointness score is zero. A virtual pass cannot repair the former word while vertex 1 remains untouched. Thus bubbling or deleting the dummy cannot be made both causal and label-preserving by retaining only a role-switch or ko marker: Proposition 39 shows that repairing the ko wall necessarily transports its labelled edge moment. The global odd flow may cancel such terms across several histories, but neither the exact draft-and-switch reduction nor a single-history dummy normalization proves the theorem.

11 The causal affine-contraction conjecture

The proof target can now be stated without Gold forms, graph enumeration, or a particular local menu.

Question 40 (global FIFO affine contraction). *Conjecture.* Fix either attacker seat and a deterministic attacker strategy S in the schedule game with one isolated dummy. Does there always exist an odd $\mathbb{F}_2$-flow through the defender branches whose terminal disjointness moment is zero, with the flow allowed to couple different defender children and different levels of the front filtration?

An affirmative answer proves the general linking theorem by Theorems 7 and 18. The tempting stronger claim—that projected continuation hulls of different defender children meet one common coset—is false. At the $P_4 + d$ state above, fix a history-dependent attacker which follows different continuations after O_2 and O_3. Under the front projection T_0 to the edge coordinates $12, 13, 23$, exact recursion gives the two open-child continuation hulls

$$\{0\}, \qquad \operatorname{Aff}\{23, 12 + 23\}.$$

They are disjoint. Exhaustive recursion over the smaller states with two real open children shows that this is the first such childwise failure.

Thus a valid contraction must transport nonzero target cosets and cancel them *across* defender children; it cannot solve every projected child as an independent zero-target subproblem. In particular, an induction classified only by dummy presence, live-set parity, attacker seat, and front position is false on reachable states. The local relations

$$s_L(d) = 0, \qquad \sum_{x \in L \cap R} s_L(x) = 0$$

provide the cone and the cut dependence, but not the required global coupling.

There is a second exact encoding of the causal obstruction. Call a complete winning line *zero-normalized* when the cumulative flip score is zero after every defender move (or forced defender pass).

Proposition 41 (response-factor normal form)**.** *First suppose the history is pass-free. Let I pair the two event occurrences O_v, C_v of each vertex, and pair each attacker event with the immediately following defender event to obtain a response matching R. If the defender moves second, $I \cup R$ is a disjoint union of alternating even cycles. If the defender moves first, it is a union of alternating cycles and one alternating path from the defender's initial OPEN to the attacker's terminal zero-charge CLOSE. Projecting R to vertex labels gives a loopless 2-regular multigraph in the second-seat case and the analogous path-plus-cycles factor in the first-seat case.*

If a forced pass occurs in the terminal singleton macro O_v, P, C_v, delete that entire macro, retain the actual charge labels on the prefix, and apply the pass-free statement on $V - \{v\}$. The vertex v is then a marked zero-charge terminal singleton outside the factor. For a second-seat defender the prefix again gives cycles. For a first-seat defender it gives cycles plus a path from the initial OPEN to the zero-charge attacker CLOSE immediately preceding the deleted macro, unless the prefix is empty, in which case only the marked singleton remains.

A response edge has zero normalization defect exactly when it is an OPEN–OPEN edge, a CLOSE–CLOSE edge whose two charges agree, or a mixed edge whose unique CLOSE has charge zero. Thus a complete zero-normalized line is equivalently a causal charge-balanced response factor of the stated type, together with the marked zero terminal singleton when a pass occurs.

Proof. Both I and, in the second-seat case, R are perfect matchings on the $2N$ event occurrences, so their union consists of alternating even cycles. In the first-seat case R leaves only the initial and terminal occurrences unmatched, producing one alternating path in addition to cycles. Every vertex has its OPEN and CLOSE occurrence in I and therefore two projected R incidences, except at those path endpoints. A projected loop would require two consecutive legal events on the same vertex, which FIFO and ko forbid. Immediately before every attacker–defender response round the cumulative score is zero. XORing the two event charges gives precisely the three cases stated. The unmatched initial event is an OPEN, and terminal victory makes the unmatched final CLOSE charge zero.

Contracting an OPEN–pass–CLOSE tail would not preserve the matching: its OPEN and CLOSE belong to the same player. Deleting all three moves instead leaves a pass-free complete prefix. If the defender moves second, both deleted touch events belonged to the attacker and the prefix matchings are perfect. If the defender moves first, both belonged to the defender; the preceding attacker CLOSE becomes the path endpoint, and its charge is zero because the following defender OPEN had to leave the cumulative score normalized. The deleted macro itself has zero charge, proving the pass case. □

The matching reformulation does not turn the causal factors into the perfect matchings of a static graph. This failure already occurs against a close-first attacker on four labels. Consider the two legal leaves

$$h_1 = O_0O_1C_0O_2C_1O_3C_2C_3,$$
$$h_2 = O_0O_2C_0O_1C_2O_3C_1C_3.$$

Their response matchings are

$$M_1 = \{O_0O_1, C_0O_2, C_1O_3, C_2C_3\},$$
$$M_2 = \{O_0O_2, C_0O_1, C_2O_3, C_1C_3\}.$$

The symmetric difference is two independent alternating four-cycles. A perfect-matching pivot may flip either cycle separately, but neither mixed factor is causal. Following the first mixed factor gives $O_0O_2C_0O_1C_2$ and then demands the response C_3 before vertex 3 has opened; the other mixed factor fails symmetrically after $O_0O_1C_0O_2C_1$. Thus compatible response factors are not exchange-closed, even inside this close-first policy.

Nor does the directed response factor determine the payoff after chronology is forgotten. The histories

$$O_0O_1O_2O_3C_0C_1C_2C_3, \qquad O_0O_1C_0C_1O_2O_3C_2C_3$$

have the same directed factor $\{O_0 \to O_1, O_2 \to O_3, C_0 \to C_1, C_2 \to C_3\}$. Taking 3 as the dummy, their real disjointness moments are respectively 0 and $02 + 12$. Consequently a factor-only edge weight cannot carry D; retaining the attacker policy and chronology restores the label but destroys alternating-cycle exchange. Ordinary Pfaffian or complementary-pivoting parity therefore cannot supply Proposition 41's missing causal extension without additional history-decorated cells.

Computational evidence. Exact minimax finds such zero-normalized strategies after adjoining the isolated dummy for every graph isomorphism class through eight real vertices, for either defender seat. In the complete order-eight census the following sharper root selectors also survive: a first-seat defender may open d; after a second-seat defender observes O_x on a real vertex, some real same-degree-parity O_y is safe; and after O_d some real OPEN is safe. These are finite certificates, not a factor-extension proof. Static routing to the dummy is already false on small reachable checkpoints: a phase pivot may first terminate at an unmatched untouched vertex.

The local choice at a normalized ordered cell makes the missing extension especially explicit. Suppose $Q = (x, y, Q_0)$, $|U|$ is even, and $\deg_U(x) = \deg_U(y) = a$. Thus both η_{xy} fibres are even. After an attacker OPEN O_z, the front charges are

$$\big(a + B(x, z),\ a + B(y, z)\big).$$

If the first bit is zero, C_x is a legal normalized phase pivot to queue (y, Q_0, z); if the bits are 11, that pivot is forbidden, while a same-η OPEN reply remains in the balanced-front corridor; and a 10 or 01 pair is a transported defect endpoint. An OPEN–OPEN reply O_w preserves equality of the eventual consecutive C_x, C_y charges exactly when $B(x+y, z+w) = 0$; further same-η pairs preserve it again. The same-curvature wall supplies an odd fan of such w, while Proposition 41 says what its one-vertex causal pivots must ultimately assemble. The unresolved theorem is precisely that every attacker-pruned partial factor admits this balanced extension; neither the cell handshake nor the static dummy endpoint proves it.

At the isolated-dummy root this can be sharpened to an odd corridor whose only uncontrolled endpoint is the dummy branch. The statement is for the stronger zero-normalized target and even real order, the case used by the exact root selectors.

Proposition 42 (least-root odd corridor)**.** *Let G have an even real vertex set R and isolated d, and assume every smaller even-real isolated-dummy board is zero-normalized safe. Fix a real opener x and put*

$$Y_x = \{y \in R - \{x\} : \deg_G(y) = \deg_G(x) \pmod 2\}.$$

Then $|Y_x|$ is odd. If every checkpoint $Q = (x, y)$ with $y \in Y_x$ is unsafe, each has an OPEN as its first spoiler. Moreover every real spoiler extends to a finite odd-augmentation corridor of unsafe same-η_{xy} OPEN pairs. The only corridor exits are

1. *consecutive CLOSEs C_x, C_y with equal charge;*
2. *a zero-charge phase pivot O_z, C_x; or*
3. *the attacker opening d.*

Summing the corridors over the odd set Y_x gives an odd aggregate exit chain.

Proof. Both degree-parity classes of an even graph have even size, by handshaking, so $|Y_x|$ is odd. At $Q = (x, y)$ the two front charges agree. If the attacker closes x, the defender closes y with the same charge and reaches exactly the smaller root on $G - \{x, y\}$ with d still isolated. Minimality makes that child safe, so the first spoiler is an OPEN.

Initially U has odd size and contains d. Equal front charges make the $\eta = 1$ fibre even; the $\eta = 0$ fibre is odd and contains d. If a real spoiler opens z, the number of remaining *real* vertices in its own η fibre is odd: in the one-fibre remove z from an even real set, while in the zero-fibre first remove d from an odd set and then remove z. Every corresponding real OPEN reply is normalized, hence unsafe under the chosen spoiler, and removes two vertices of one fibre. It preserves equal front charges, odd $|U|$, and the untouched dummy, so the construction iterates. Each round removes two real untouched vertices. It must therefore terminate at a CLOSE–CLOSE exit, at an OPEN after which C_x has charge zero, or when the attacker selects d. Every internal fan and the outer Y_x fan have odd augmentation, proving the last assertion. $\square$

The proposition isolates a precise parity leak rather than closing the root induction. After O_d there are $|R| - 2$ real OPEN replies, an even fan, and C_x is an additional normalized reply exactly when the common initial front charge is zero. Exact minimax through the complete eight-real census finds that O_d is never a spoiler at an unsafe same-parity root, but no fan-level proof is known. Pairwise dummy diamonds are insufficient: the checker has a root for which a real-spoiler child crossed with d is itself unsafe, although other real replies make the actual O_d branch safe.

One unbounded attacker class can nevertheless be contracted completely. It is the class used below to generate the causal simplex caps, so the result is not a bounded-support statement in disguise. Call an attacker *close-first* if it closes the front whenever that is legal and opens only when ko or an empty queue forbids a close. Write J_R for the all-edge chain on the real vertices, and for an ordering P of $R \sqcup \{d\}$ write P_R and P_R^2 for the real-edge traces of the path and its square.

Theorem 43 (complete close-first contraction)**.** *For every number of real vertices, every close-first attacker admits an odd compatible response chain with zero terminal disjointness moment. This holds whether the attacker occupies the first or the second seat.*

Proof. Suppose first that the attacker moves first and opens a real vertex p. The defender opens v_2 and thereafter, following every attacker close, opens an arbitrary untouched vertex; after the last opening the remaining fronts close. Every Hamiltonian ordering

$$P = (p, v_2, \ldots, v_N)$$

is therefore compatible. Its overlap graph is the path P_R, so

$$D_R = J_R + P_R. \tag{50}$$

The affine hull of these path traces is the whole $E(K_R)$. Indeed, if a real edge weighting A has constant path sum, compare two orders which are identical except for the local segments

$$a, d, x, b, \qquad a, x, d, b.$$

Their difference gives $A_{xb} = A_{ax}$. Varying the three real vertices makes every real edge carry one common value t. Putting the dummy at an endpoint or in the interior changes the number of real path edges by one, so $t = 0$. The cases with at most three real vertices follow directly; for three, the three traces from

$$(p, x, y, d), \quad (p, y, x, d), \quad (p, d, x, y)$$

XOR to J_R. Hence J_R is an odd affine sum of the P_R, and (50) gives a zero sum of the corresponding D_R.

If $|R| \leq 1$, a dummy-root game is trivial. Otherwise, if the first attacker opener is the dummy, use two compatible response families. One-block play realizes every Hamiltonian path on R as the real overlap graph. Alternatively, for each real x the defender plays

$$O_d, O_x, C_d, C_x.$$

The queue is empty, so the attacker must open some history-dependent $y_x \in R - \{x\}$; resuming one-block play realizes an isolated x together with every Hamiltonian path on $R - \{x\}$ beginning at y_x. Any edge weighting constant on all Hamiltonian paths of K_R is uniform: comparing the paths x, a, y, r, L and y, a, x, r, L gives $A_{yr} = A_{xr}$, and varying the vertices makes $A_e = t$ for every real edge when $|R| \geq 4$; equality of the three two-edge path sums gives the same conclusion when $|R| = 3$. A full path and an isolate-first path have respectively $|R| - 1$ and $|R| - 2$ edges, so constancy on their union forces $t = 0$. Thus this union has full affine hull and again contains J_R. For $|R| = 2$ the two families have traces J_R and 0. This proves the first-seat assertion, including the dummy-root wall which one-block paths alone would not cover.

Now let the close-first attacker move second. The case $|R| \le 1$ is trivial, so assume at least two real vertices. The defender first opens a vertex p; ko forces the attacker to open a vertex $q(p)$; the defender then always opens while an untouched vertex remains. Every remaining opening order is under defender control, and the real overlap graph is P_R^2 for a Hamiltonian order with prefix $(p, q(p))$. Hence

$$D_R = J_R + P_R^2. \tag{51}$$

It remains to show that J_R lies in the affine hull of the available square traces.

First suppose some real p has real $q = q(p)$. Restrict to that root. If $F_A(P) = \langle A, P_R^2 \rangle$ is constant on all orders with prefix (p, q), extend the real-edge weighting A by zero on dummy edges. Swapping adjacent tail vertices in the local words

$$a, b, x, y, c, e, \qquad a, b, y, x, c, e$$

gives

$$A_{ax} + A_{ye} + A_{ay} + A_{xe} = 0. \tag{52}$$

For $|R| \ge 7$, putting $e = d$ and varying the tail shows that all p-tail edges have one value α, all q-tail edges one value β, and all tail-tail edges one value t; write $A_{pq} = \gamma$. Moving d to tail positions $3, 4, N-1, N$ changes the nonconstant part of F_A through

$$\gamma + \beta + t, \quad \gamma + \alpha + \beta, \quad \gamma + \alpha, \quad \gamma + \alpha + t.$$

Constancy forces $\alpha = \beta = t = 0$. The only possible residual functional is supported on pq; its constant value is $\gamma = \langle A, J_R \rangle$. By affine separation this is exactly the condition $J_R \in \mathrm{Aff}\{P_R^2 : P \text{ begins } (p, q)\}$. The finitely shorter ranks are witnessed directly by the odd supports of sizes $1, 1, 3, 5, 7$ for $|R| = 2, 3, 4, 5, 6$; these explicit orders are checked in the companion experiment by `close_first_low_rank_certificates`.

It remains that $q(p) = d$ for every real root p. Include also the dummy root, where the forced second opener is some real q_0. Thus the response family contains all square paths beginning (p, d), for every real p, together with all those beginning (d, q_0). If F_A is constant on this union, comparing the two allowed prefixes $(d, q_0, x, y, \ldots)$ and $(q_0, d, x, y, \ldots)$ gives $A_{q_0 y} = 0$. Further adjacent-tail comparisons are also explicit. For distinct $p, x, y, z \in W = R - \{q_0\}$, compare

$$(p, d, x, q_0, y, z, \ldots), \qquad (p, d, q_0, x, y, z, \ldots).$$

The difference is $A_{px} + A_{xz}$, so every edge inside W has one value t. On $(d, q_0, w_1, \ldots, w_{|W|})$ the W-edge contribution is $(2|W| - 3)t = t$, whereas on an order $(p, d, x, \ldots, q_0)$ with q_0 last and all of $W - \{p, x\}$ in the omitted positions, it is $t + (2(|R| - 2) - 3)t = 0$. Both values equal the common constant, so $t = 0$. Hence $A = 0$, and the union in fact has full affine hull. Direct odd supports of sizes $1, 1, 3$ cover $|R| = 2, 3, 4$ in the same companion function; the comparison applies from $|R| = 5$ onward. Therefore J_R is an odd affine sum in every second-seat case, and (51) finishes the proof. $\square$

Theorem 43 closes the policy responsible for the unbounded caps, but not an arbitrary policy. It also exposes why a simple monotonicity argument fails: replacing one forced close by an attacker OPEN prunes the very adjacent-event diamond which would add the correcting rooted triangle. A responder restricted to OPEN while untouched vertices remain is already false on $K_4 - e$ plus the dummy; the winning unrestricted reply is a phase-changing front CLOSE. Thus neither "more overlap is easier" nor a single-block reduction removes the causal phase problem.

Positional reduction makes the last attempted normalization exact. Let a *leafmost deviation* be an attacker state $s = (U, Q, \mathsf{ko} = 0, t)$, with $Q = (f, Q_0)$, at which a positional winning policy chooses O_z although C_f is legal, and is close-first at every strict descendant. Suppose replacing that move by C_f, while retaining the positional tail, is losing.

Proposition 44 (leafmost-deviation spoiler dichotomy)**.** *A defender spoiler after the replacement C_f has exactly one of the following forms.*

1. *It is O_z and $Q_0 \neq ()$. Then the two paths $O_z; C_f$ and $C_f; O_z$ reconverge publicly, and irreducibility forces $B(f, z) = 1$.*
2. *It is O_z and $Q_0 = ()$. The two paths agree except that the latter has a ko-protected singleton queue (and their score sheets may differ by $B(f, z)$); this is the singleton wall.*
3. *It is O_w for $w \neq z$, producing genuinely different queue orders.*
4. *It is the close of the new front g, when $Q_0 = (g, Q_1)$; this is the phase-changing two-close spoiler.*

A pass cannot be the spoiler.

Proof. After C_f, the vertex z is still untouched, so an OPEN is legal and a pass is not. The displayed OPENs and, when Q_0 is nonempty, the close of its front exhaust the legal moves. If the spoiler is O_z away from the singleton wall, Proposition 31 says that its state and the winning $O_z; C_f$ state differ only by the score bit $B(f, z)$. If that bit were zero, positionality would give the same winning continuation after the replacement, a contradiction. The remaining cases are the three exact ways public reconvergence fails: ko at a singleton, a different append/delete order, or an additional front close. □

The untouched-dummy part of the conditioned tail does admit a complete contraction, for an arbitrary preloaded queue rather than only a root.

Theorem 45 (untouched-dummy close-first tail)**.** *Let the defender move from a state with nonempty queue $Q = (q_1, \dots, q_L)$ and untouched set U containing the isolated dummy d. If the attacker is close-first from this state onward, the defender can force future score zero.*

Proof. The defender may choose an arbitrary ordering $P = (p_1, \dots, p_m)$ of U, always OPENing the next p_i. While untouched vertices remain, the close-first attacker closes one FIFO front after each such OPEN. Once U is empty all later charges vanish. Directly reading the front at each round gives the future score

$$F(P) = \sum_{\substack{i<j \\ i \leq L}} B(q_i, p_j) + \sum_{a+L<b} B(p_a, p_b). \tag{53}$$

It suffices to show that $F(P) = 0$ for some permutation.

Suppose instead that F is constant. Swap adjacent entries x, y in positions $j, j+1$. The only changed contributions are those at distance one across the queue lag, and

$$F(\cdots xy \cdots) + F(\cdots yx \cdots) = B(x + y, a_j + b_j), \tag{54}$$

where

$$a_j = \begin{cases} q_j, & j \leq L, \\ p_{j-L}, & j > L, \end{cases} \qquad b_j = \begin{cases} p_{j+L+1}, & j + L + 1 \leq m, \\ 0, & j + L + 1 > m. \end{cases}$$

Put $y = d$ in (54); every pairing with d vanishes.

If $m \le L+1$, every b_j vanishes and the equations give $B(q_j, x) = 0$ for every queue row which occurs in the first sum of (53); the second sum is empty. Hence $F = 0$. If $m = L+2$, use the final swap $j = L+1$: here $a_j = p_1$ and $b_j = 0$, and arbitrary placement of the other vertices proves $B|_U = 0$. The swaps $j \le L$ then kill every scoring q_j row. If $m \ge L+3$, the final swap again has $a_j \in U, b_j = 0$ and proves $B|_U = 0$; the remaining swaps kill the scoring queue rows in the same way. Thus a constant F is the constant zero. If F is nonconstant, its $\mathbb{F}_2$-valued image already contains zero. In either case choose a zero-valued ordering P and follow the displayed OPEN strategy. $\square$

The scalar conclusion hides a stronger affine fact. For disjoint vertex sets write

$$\mathcal{E}(S) = \operatorname{span}_{\mathbb{F}_2}\{st : s, t \in S,\ s \neq t\}, \qquad \mathcal{E}(S,T) = \operatorname{span}_{\mathbb{F}_2}\{st : s \in S,\ t \in T\}.$$

Theorem 46 (untouched-dummy close-first full live face)**.** *In the state of Theorem 45, write $U = \{d\} \sqcup R$ and suppose ko is clear. For the fixed ordinary close-first attacker, let W be the even continuation-moment space, projected to the coordinates which are still live at this state. Then*

$$W = \mathcal{E}(R) \oplus \mathcal{E}(Q,R). \tag{55}$$

Equivalently, the affine hull of the future terminal moments is the whole live face.

Proof. Put $X = \mathcal{E}(R) \oplus \mathcal{E}(Q,R)$. It suffices to show that every linear functional ℓ on X which annihilates W is zero. Such an ℓ has one constant value c on all odd terminal continuation moments, since the difference of two such moments lies in W. We induct on $|R|$, uniformly in the nonempty queue $Q = (f, Q_0)$.

Assume first that $|R| \ge 2$. For each $x \in R$, take the defender move O_x; the close-first attacker answers C_f. The child is again a clear defender state with untouched dummy and real untouched set $R - \{x\}$. An even chain inside this child is also an even chain at the parent, because its common prefix has even total coefficient. By induction the child even space is its whole live coordinate space. Therefore ℓ vanishes on

$$\mathcal{E}(R - \{x\}) \oplus \mathcal{E}(Q_0, R - \{x\}) \oplus \mathcal{E}(\{x\}, R - \{x\}).$$

As x varies, this kills every coefficient in $\mathcal{E}(R)$ and in $\mathcal{E}(Q_0, R)$. The only coefficients which may remain are those of fR. The charge of the forced C_f in the O_x branch then gives

$$c = \sum_{u \in R - \{x\}} \ell(fu) \qquad (x \in R). \tag{56}$$

If the defender instead begins with C_f, every later live coordinate has already been killed, while this close contributes all of fR. Hence

$$c = \sum_{u \in R} \ell(fu). \tag{57}$$

Comparing the two displays gives $\ell(fx) = 0$ for every x, and then $c = 0$.

If $R = \varnothing$, then $X = 0$. Finally let $R = \{u\}$ and write $Q = (v_0, \ldots, v_{L-1})$. The defender continuation beginning with O_u has zero live moment, so $c = 0$. For each $0 \le k < L$, there is also a legal continuation in which precisely $v_0, \ldots, v_k$ close before O_u: if k is even, begin

$$O_d, C_{v_0}, C_{v_1}, \ldots, C_{v_k}, O_u,$$

and if k is odd, omit the initial O_d. In the first word the attacker takes the even-indexed closes and the defender the odd-indexed ones; in the second word these roles are reversed. Thus every attacker move displayed is its forced close-first reply, every displayed defender close is legal, and the last close is followed by the defender's O_u. After this prefix complete the play arbitrarily. Constancy of ℓ now gives

$$\sum_{i=0}^{k} \ell(v_i u) = 0 \qquad (0 \leq k < L).$$

Successive differences kill every $\ell(v_i u)$. Thus the annihilator of W in X^* is zero in every case, proving (55). Since the odd continuation set is a nonempty W-coset contained in X, it is then all of X as well. □

The word "live" is essential. Coordinates frozen by the prefix are not in X and need not be absorbable by the tail. With sole real edge ab and c, d isolated, the legal prefix

$$O_a, O_c, C_a, O_b$$

reaches $Q = (c, b)$ and $U = \{d\}$ with accumulated moment ab. Every future charge is zero, so Theorem 46 correctly gives $W = X = 0$ but cannot remove the frozen prefix bit. This is precisely why the conditioned tail theorem does not by itself close the root ancestry.

Consequently, while the dummy is untouched a leafmost deviation can only enter a close-first tail on residual target zero. In the minimum-bad descent, residual one must first cross to zero through the preceding odd CLOSE; if the dummy has already been spent, Theorem 45 no longer applies. These are exactly the odd-CLOSE spike and post-dummy domination boundaries in Proposition 13.

The dummy hypothesis can in fact be removed completely once the tail begins at a clear defender checkpoint. This is strictly stronger than the root close-first contraction: the queue may be arbitrary, and the conclusion is a single scalar response rather than an affine combination of root histories.

Theorem 47 (conditioned close-first contraction)**.** *Let the defender move from a coherent state with nonempty queue,* $\mathsf{ko} = 0$*, and arbitrary untouched set* U*. Against an attacker which CLOSEs whenever CLOSE is legal, the defender has a continuation of future score zero. No dummy or parity hypothesis on the live graph is required.*

Proof. Fix a close-first attacker and suppose that it forces future score one from some such defender state. Among coherent ko-clear defender states with nonempty queue carrying such a target-one close-first strategy tree, choose a counterexample of minimum remaining move rank. Legal moves preserve coherence and strictly decrease this rank. Write

$$Q = (f, Q_0).$$

If U is empty, every remaining charge is zero. If $U = \{x\}$, the defender opens x; the forced C_f and every later close then see empty untouched set. Hence $|U| \geq 2$.

We first dispose of the only small fibre which is not covered by the uniform degree comparison below. Let $U = \{x, y\}$ and write $Q = (f, Q_0)$. If f misses, say, y, the defender opens x. The forced C_f sees only y and has charge zero; the defender then opens y, after which all charges vanish. We may therefore assume that f sees both x and y.

After O_x, C_f the score has changed by one and the defender faces the singleton untouched set $\{y\}$ with queue (Q_0, x). It may simply drain that queue. The close-first attacker also closes whenever it moves, and after the queue empties the sole remaining vertex opens, passes if necessary, and closes at charge zero. Since the attacker tree forces the score already present after C_f, this defender line gives

$$\sum_{v\in Q_0} B(v,y) + B(x,y) = 0.$$

Reversing x, y gives

$$\sum_{v\in Q_0} B(v,x) + B(x,y) = 0.$$

Now instead drain the original queue before either untouched vertex opens. Its total charge is

$$B(f,x) + B(f,y) + \sum_{v\in Q_0} \big(B(v,x) + B(v,y)\big) = 0:$$

the front contribution is $1+1$, and the two displayed equations leave two copies of $B(x, y)$. Once the queue is empty, the first remaining OPEN creates ko and forces the other vertex to OPEN before either can close, so all later charges vanish. This zero-charge defender continuation contradicts the target-one tree. Thus $|U| = 2$ cannot be a counterexample.

Assume now $|U| \geq 3$. For every $x \in U$, let the defender open x. The attacker must close f, reaching a smaller clear defender state with nonempty queue and prefix charge

$$c_x = \deg_{U-\{x\}}(f).$$

If some c_x were zero, the restricted attacker tree would be a smaller counterexample. Hence $c_x = 1$ for every x. With $p = \deg_U(f)$ this says

$$p + B(f,x) = 1 \qquad (x \in U).$$

The alternative $p = 1$ would make every $B(f, x)$ zero and hence $p = 0$. Therefore f is universal on U, $|U|$ is even, and $p = 0$; in particular $|U| \geq 4$.

Suppose first that $Q = (f)$. The defender closes f at charge zero. At the now empty, ko-clear queue the attacker opens some z, creating the ko-protected singleton queue (z); the defender opens an arbitrary $y \neq z$, and the attacker is forced to close z. Minimality would require

$$\deg_{U-\{z,y\}}(z) = 1 \qquad (y \neq z),$$

since a zero charge would expose a smaller counterexample. Write

$$a_y = B(z,y), \qquad \delta = \sum_{u\neq z} B(z,u).$$

The displayed equations are $\delta + a_y = 1$ for every y, so all a_y have one value b. But $|U| - 1$ is odd, whence $\delta = b$ and every displayed charge is actually zero, a contradiction.

Finally write $Q = (f, h, Q_1)$. For distinct $x, y \in U$, use the four-move prefix

$$O_x, C_f, O_y, C_h.$$

The first close has charge one. If $e_{xy} = \deg_{U-\{x,y\}}(h)$ were also one, the two charges would cancel and leave a smaller counterexample. Thus $e_{xy} = 0$ for every pair. Comparing two pairs with one common vertex shows that $B(h, -)$ is constant on U; since $|U|$ is even, $\deg_U(h) = 0$. The defender can now play C_f and the attacker must answer C_h; both charges are zero. If Q_1 is nonempty this is immediately a smaller counterexample. If Q_1 is empty, the defender opens some x onto the empty queue, creating ko; the attacker is forced to open some $y \neq x$. The resulting clear state has queue (x, y), the defender to move, and smaller rank, so it is again a counterexample. This final contradiction proves the theorem. $\square$

The preceding theorem has a useful stopped extension at an empty root. The notation below is deliberately different from the universal-full-fan predicate $P(H)$ used earlier in the repair-forest discussion.

Call an attacker *stopped close-first* if its rules are as follows. At a clear node—that is, $Q \neq ()$ and $\mathsf{ko} = 0$—it must CLOSE when the current score is zero, while at score one it may STOP and win immediately; if it declines to STOP, it must still CLOSE. When ko or an empty queue forbids CLOSE, it may choose any legal OPEN; at a nonterminal state with no legal move, the usual forced pass occurs. The defender retains every ordinary legal move. A completed play is won by the attacker exactly when its terminal score is one. Write $\mathsf{P}_{\text{stop}}(H)$ when the defender wins this stopped game from score zero, empty queue, and all vertices of H untouched, with the defender to move. No dummy is assumed.

For distinct $x, y \in V(H)$, call the ordered pair $x \to y$ *bad* when the stopped attacker wins from the defender state

$$U = V(H) - \{x, y\}, \qquad Q = (x, y), \qquad \mathsf{ko} = 0, \qquad g = 0.$$

All quantities in the next lemma lie in $\mathbb{F}_2$. Put

$$X_z = B(x, z), \quad Y_z = B(y, z), \quad a = B(x, y),$$

$$\alpha = \sum_{z \in U} X_z, \qquad \beta = \sum_{z \in U} Y_z,$$

and, for the full graph H,

$$p_v = \deg_H(v), \qquad b_v = \sum_{w \sim v} p_w, \qquad r = \alpha + \beta = p_x + p_y, \qquad \epsilon = |V(H)|.$$

Lemma 48 (stopped bad-pair fan)**.** *Assume $\mathsf{P}_{\text{stop}}(K)$ for every proper induced subgraph K of H, and assume the present lemma for every graph of smaller order. If $x \to y$ is bad in H, then*

$$\beta = 1, \qquad b_x = 1, \qquad \alpha = 1 \Longrightarrow \epsilon = 1. \tag{58}$$

Proof. If $\alpha = \beta = 0$, the defender plays C_x, the score-zero attacker is forced to play C_y, and the game returns at score zero to the smaller empty root on U. Thus

$$\alpha = 0 \Longrightarrow \beta = 1. \tag{59}$$

There is a second elementary absorber. If some $z \in U$ satisfies

$$p_z = r, \qquad X_z + Y_z = r, \tag{60}$$

then the defender uses

$$O_z, C_x, C_y, C_z,$$

where the first and third moves are its own. The first two close charges agree because

$$\alpha + X_z = \beta + Y_z,$$

and the last charge is $p_z + X_z + Y_z = 0$. The line therefore returns at score zero to the smaller empty root on $U - \{z\}$. Hence the fibre (60) is empty for a bad pair.

The parity of that fibre is the sum over U of

$$(p_z + 1 + r)(X_z + Y_z + 1 + r).$$

Now

$$\sum_U p_z = \sum_U (X_z + Y_z) = r$$

and

$$\sum_U p_z(X_z + Y_z) = b_x + b_y + ar.$$

Since $|U| = \epsilon$ modulo two, emptiness gives the exact moment identity

$$b_x + b_y = (1 + r)\epsilon + ar. \tag{61}$$

It remains to use the complete zero-charge OPEN fan. Set

$$Z = \{z \in U : X_z = \alpha\}, \qquad c = \sum_{w \in U} X_w Y_w.$$

For every $z \in Z$, the defender move O_z followed by the forced C_x has charge zero and reaches the smaller bad pair $y \to z$ in $H - x$. Applying the inductive lemma to this child gives

$$p_z + \alpha + Y_z = 1 \qquad (z \in Z), \tag{62}$$

because the left side is its second-front charge. The child's source bit is

$$b_y^{H-x} = b_y + ap_x + c = 1. \tag{63}$$

Indeed, deleting x changes the degree bit of every remaining w by $B(x, w)$. Finally the child front charge is $\alpha'_z = \beta + Y_z$. If $\epsilon = 1$, then $H - x$ has even order, so the inductive implication in (58) forces

$$\epsilon = 1 \Longrightarrow Y_z = \beta \qquad (z \in Z). \tag{64}$$

First suppose $Z = \varnothing$. Then $\alpha = 0$: if $\alpha = 1$, its odd set of neighbours in U would be nonempty. Since no X_z is zero, x is universal on U, and $0 = \alpha = |U| = \epsilon$. Equation (59) gives $\beta = 1$. Hence

$$p_x = a, \qquad p_y = a + 1, \qquad \sum_U p_z = p_x + p_y = 1.$$

Because x is universal on U,

$$b_x = ap_y + \sum_U p_z = a(a + 1) + 1 = 1.$$

Thus (58) holds at every terminal domination wall.

Assume now that $Z \neq \varnothing$ and $\alpha = 1$. The set $Z = N_H(x) \cap U$ has odd cardinality, and (62) says $p_z = Y_z$ on Z. Therefore

$$b_x = ap_y + \sum_Z p_z = a(1 + \beta) + c = ar + c. \tag{65}$$

Also $p_x = a + 1$, so $ap_x = 0$, and (63) gives $b_y = 1 + c$. Substitution in (61) yields

$$1 + ar = (1 + r)\epsilon + ar,$$

or $\beta\epsilon = 1$, since $r = 1 + \beta$. Thus $\beta = \epsilon = 1$. Equation (64) now gives $Y_z = 1$ for every $z \in Z$. Hence $c = 1$, while $r = 0$, and (65) gives $b_x = 1$.

Finally suppose $Z \neq \varnothing$ and $\alpha = 0$. By (59), $\beta = r = 1$. Here Z is the set of nonneighbours of x in U, and $|Z| \equiv \epsilon \pmod 2$. Summing (62), now $p_z + Y_z = 1$, over Z gives

$$b_x + c = \epsilon. \tag{66}$$

Explicitly,

$$\sum_Z p_z = \sum_U (1 + X_z) p_z = r + b_x + a p_y = 1 + b_x$$

because $p_y = a + 1$, while $\sum_Z Y_z = \beta + c = 1 + c$. Since $p_x = a$, equation (63) gives $b_y = 1 + a + c$. Equation (61), with $r = 1$, gives $b_x + b_y = a$, hence

$$b_x + c = 1.$$

Comparison with (66) yields $\epsilon = 1$. Equation (64) forces $Y_z = 1$ on Z. Thus $\sum_Z Y_z = 1$; since $\sum_U Y_z = \beta = 1$, we have $c = 0$, and therefore $b_x = 1$. This proves the lemma. $\square$

Theorem 49 (stopped close-first empty-root contraction)**.** *For every finite graph H, $\mathsf{P}_{\mathrm{stop}}(H)$ holds. Equivalently, from a defender-first empty no-dummy root of score zero, no stopped close-first attacker can force either a score-one STOP or terminal score one.*

Proof. We prove the theorem and Lemma 48 simultaneously by strong induction on $|V(H)|$: the lemma at order n uses the theorem and the lemma only on smaller induced graphs, and then proves the theorem at order n.

The empty graph is terminal at score zero. On a one-vertex graph the moves are an OPEN, a forced pass clearing ko, and a zero-charge CLOSE. Now suppose $|V(H)| \geq 2$ and that $\mathsf{P}_{\mathrm{stop}}(H)$ fails. After any first defender OPEN O_x, ko forbids CLOSE and the attacker must have some winning OPEN O_y. Thus every vertex of H has an outgoing bad arc $x \to y$.

Give each vertex the rank

$$R(v) = \begin{cases} 2, & (p_v, b_v) = (0, 1), \\ 1, & (p_v, b_v) = (1, 1), \\ 0, & b_v = 0. \end{cases}$$

Every bad arc strictly decreases this rank. If $\alpha = 0$, then $\beta = 1$, $a = p_x$, and $p_y = 1 + p_x$. Equation (61), with $r = 1$ and $b_x = 1$, gives $b_y = 1 + p_x$. Hence

$$(0,1) \longrightarrow (1,1) \quad \text{or} \quad (1,1) \longrightarrow b = 0.$$

If $\alpha = 1$, Lemma 48 gives $\beta = \epsilon = 1$. Thus $r = 0$, $p_y = p_x$, and (61) gives $b_y = 0$.

A finite directed graph in which every vertex has positive outdegree contains a directed cycle, while strict descent of R forbids one. This contradiction proves $\mathsf{P}_{\mathrm{stop}}(H)$ and completes the mutual induction. $\square$

Theorem 49 closes exactly the empty-root base left by a STOP normalization. It does not justify iterating a branchwise collapse through the odd-CLOSE spike (A) or the singleton ko wall (B′) of Proposition 13. There the closest $C_y; O_x$ and $O_x; C_y$ histories differ by the frozen edge xy and by ko until another real OPEN occurs; descendant close-first spaces cannot in general absorb that defect. The unresolved bridge is therefore an odd affine cancellation across the complete defender fan at the (A)/(B′) ancestor, not another empty-root theorem.

In particular, the poststate of a leafmost attacker OPEN is a clear defender state with nonempty queue and a close-first attacker tail. Such a deviation can force only residual target zero, whether or not the dummy has already been spent. The remaining root problem is therefore not the conditioned tail conjecture stated in the previous draft: it is the transport of those target-zero deviations across the preceding odd CLOSE.

Neither "the dummy has already been touched" nor "z is the unique untouched neighbour of f" follows locally. For the latter, take $K_{2,2}$ on parts $\{0,1\},\{2,3\}$ after the reachable spent-dummy prefix $O_dO_1O_0C_d$: at $U=\{2,3\},Q=(1,0)$ the irreducible O_2 has a close-first zero tail although front 1 has two untouched neighbours. If the defender opens 3, the untouched set is empty; if it closes 1, the forced close of 0 has the same unit charge, and all later charges vanish. Replacing O_2 by C_1 is spoiled by O_2,O_3, leaving the forced C_0 as the sole unit charge. For the former claim, with sole real edge 01, the reachable prefix $O_2O_1C_2$ leaves $U=\{0,d\},Q=(1)$; after O_0, either O_d or C_1 leaves only zero charges, while replacing it by C_1 has unit charge and the spoiler O_d makes every later charge zero. Thus the isolated dummy remains untouched in this irreducible target-zero example. Hence the last-deviation program does not reduce to a local uniqueness lemma. Theorem 47 closes every post-OPEN close-first tail, but only at residual target one; the irreducible examples above live on residual target zero after an earlier odd CLOSE.

This distinction cannot be removed by requiring close-first play only on the residual-one sheet. On the graph

$$E=\{01,02,12,03\}$$

consider the clear defender state

$$U=\{0,2\},\qquad Q=(1,3),$$

with residual target one. There is an odd-forcing attacker policy which always CLOSEs on residual one and makes exactly one voluntary target-zero OPEN. After defender O_0 it plays the unit C_1; the branches O_2 and C_3 are answered respectively by the zero C_3 and by O_2 in place of the now-odd C_0. The O_2 branch is symmetric, except that defender C_3 has charge one and is followed by the unit C_2. If the defender initially plays the zero C_1, the attacker plays the unit C_3; the two remaining vertices then open across the empty-queue ko wall and all later charges vanish. Every defender line has future score one. The sole deviation is the target-zero singleton state $U=\{2\},Q=(0)$, where replacing O_2 by the unit C_0 is spoiled by the ko-wall O_2. An isolated dummy may already have been spent along the legal zero prefix O_d,O_1,O_3,C_d. Thus induction on the last residual switch is false at a reachable conditioned state; the root ancestry must still couple different earlier defender branches.

This boundary also explains why the nearest imported techniques do not close the problem. A standard strategy-stealing pass must be optional and harmless; the dummy has two mandatory events which change FIFO legality. A static copycat argument requires an involution or a parity-cell partition, absent on the 64 finite witnesses above. Local complementation changes the required pairing data but is not a legal FIFO move. Queue layouts supply terminology for nonnesting but no adversarial parity strategy.

Computational evidence. A useful stronger conjecture is false: without the dummy, the initial mover does not always force even score on even-order graphs. The first failures occur at order eight, in exactly seven isomorphism classes: `GCZN^{`, `GCrU~{`, `GEhvn{`, `GCx}~{`, `GEjt~{`, `GEl~v{`, and `GUZv^{`. They are anti-mover-controlled: the second player can force the target parity, whether that target is even or odd. In particular the weaker finite conjecture that a second-seat even player wins every even-order no-dummy board survives this screen, but it is not proved. Bare alternating rank or Witt class cannot classify these outcomes: P_3 and $K_2 \sqcup K_1$ have congruent rank-two alternating adjacency forms over $\mathbb{F}_2$ but different seat outcomes.

The smallest genuine affine cancellation is already a three-history fan, not a dummy cone. With real vertices $1, 2, 3$, let a first-seat attacker open the dummy and thereafter close whenever possible, otherwise opening the least untouched vertex. After the shared prefix O_d, O_1, C_d, the defender can realize the three compatible terminal histories

	remaining moves	$D(h)$
(i)	O_2, C_1, O_3, C_2, C_3	$\{13\}$
(ii)	O_3, C_1, O_2, C_3, C_2	$\{12\}$
(iii)	C_1, O_2, O_3, C_2, C_3	$\{12, 13\}$.

Their three vectors XOR to zero, although none is zero. In (iii), O_2 opens an empty queue and O_3 is the ko-forced response, so all three rows are legal. The cancellation happens after the dummy has closed.

Computational evidence. Three histories do not suffice uniformly. With five real vertices plus d, let a first-seat attacker close whenever legal and otherwise open the least untouched vertex. Its 132 distinct response scores contain neither zero nor three vectors with XOR zero. The minimum odd certificate has support five; one is

$$\begin{aligned}
&\{01, 02, 03, 13, 14, 24\}, \qquad &&\{01, 02, 04, 12, 13, 23, 24\},\\
&\{01, 03, 04, 12, 14, 23\}, \qquad &&\{02, 03, 04, 12, 14, 23\},\\
&\{01, 02, 03, 04, 12, 14, 23\}.
\end{aligned}$$

The five rows XOR to zero. Thus the triangular fan is a real local cancellation, not a bounded global normal form for affine response flows.

The failure of bounded support persists after quotienting by cuts, and is not an isolated finite-search phenomenon.

Proposition 50 (causal simplex caps)**.** *For every $r \geq 4$ there is a reachable defender node below a fixed first-seat attacker at which the terminal cut-quotient labels project onto*

$$\mathcal{C}_r = \{\mathbf{1}\} \cup \{\mathbf{1} + e_i : i = 1, \ldots, r\} \subseteq \mathbb{F}_2^r.$$

The set $\mathcal{C}_r$ contains neither zero nor three distinct elements whose sum is zero. Moreover any odd terminal response chain with cut-valued moment has support at least r when r is odd and at least $r + 1$ when r is even. These bounds are unbounded in r. At $r = 4$ the bound five is attained by an explicit compatible response chain.

Proof. Take real vertices $0, v$ and an r-set U, together with the isolated dummy d. Let the first-seat attacker and defender follow the prefix

$$A : O_0, \quad D : O_d, \quad A : C_0, \quad D : O_v, \quad A : C_d.$$

The accumulated disjointness chain is $\sum_{x \in U \cup \{v\}} 0x = \delta\{0\}$, hence vanishes in the cut quotient, and the resulting state is $Q = (v)$, with untouched set U and the defender to move. From then on let the attacker close whenever possible, otherwise open the least untouched vertex.

Project $T_0 D$ onto the star coordinates vi, $i \in U$. If the defender first closes v, its interval is disjoint from every vertex of U, giving $\mathbf{1}$. If the defender first opens i, the attacker immediately closes v, so v overlaps only i and the label is $\mathbf{1} + e_i$. Later play cannot change these coordinates. Every listed first move is legal, so the projected terminal label set is exactly $\mathcal{C}_r$.

Put $a = \mathbf{1}$ and $b_i = \mathbf{1} + e_i$. A relation

$$\alpha a + \sum_i \beta_i b_i = 0$$

has, on coordinate j, the equation $\alpha + \sum_i \beta_i + \beta_j = 0$. Hence all β_j equal one bit t. For $t = 0$ the relation is trivial; for $t = 1$ it is the unique nonempty relation, with

$$\beta_i = 1 \text{ for every } i, \qquad \alpha = 1 + r \pmod 2.$$

Its support is therefore r for odd r and $r+1$ for even r, in either case odd. Since any zero quotient chain projects to a relation in $\mathcal{C}_r$, this proves the support lower bound. For $r \geq 4$, sums of two distinct elements of $\mathcal{C}_r$ have weight one or two, while its elements have weight $r-1$ or r; this proves the cap assertion.

For attainment at $r = 4$, label the untouched vertices $1, 2, 3, 4$, take $v = 5$ and $d = 6$, and retain the displayed prefix. The following five defender continuations are compatible with the close-first attacker:

$$\begin{array}{r|l} 766 & O_3C_5C_3O_1O_2C_1C_2O_4PC_4 \\ 797 & O_2C_5O_4C_2O_3C_4O_1C_3C_1 \\ 853 & O_1C_5O_3C_1O_4C_3O_2C_4C_2 \\ 861 & C_5O_1O_3C_1O_4C_3O_2C_4C_2 \\ 491 & O_4C_5O_1C_4C_1O_2O_3C_2C_3. \end{array}$$

Here the left column is the binary integer for T_0D in lexicographic edge order

$$12, 13, 14, 15, 23, 24, 25, 34, 35, 45.$$

Directly,

$$766 \oplus 797 \oplus 853 \oplus 861 \oplus 491 = 0.$$

The schedules and labels are independently checked by `experiments/linking_game.py::close_first_schedule_quotient`. The lower bound already proved makes five the exact minimum. □

Proposition 50 rules out more than a universal three-history fan. No induction can bound the support of a certificate inside each defender child independently. A root contraction must couple these growing caps to a wider collection of earlier sibling branches.

Nor is one arbitrary sibling enough. *Computational evidence.* At the immediately preceding checkpoint

$$U = \{1, 2, 3, 4, 5\}, \qquad Q = (6),$$

with 6 the dummy and the defender to move, restrict attention to the two defender children O_1 and O_5. Exact backward induction constructs a history-dependent attacker whose combined terminal image in the T_0 quotient is a 52-point cap: zero is absent and no two distinct labels have their sum in the image. The cap and the independent target-safety recursion are pinned by `experiments/linking_game.py::quotient_target_response_labels`. Thus the proposed ancestor rule "every cap child is broken by each additional sibling" is false already at $r = 4$. This is not a root counterexample: the full defender fan also contains C_6, O_2, O_3, O_4, and those branches remain available to a global odd flow. It is not an affine counterexample even on the partial fan, since the cap itself contains the five-circuit $7 \oplus 11 \oplus 13 \oplus 14 \oplus 15 = 0$.

The exact ancestor-cap escape rule is false as well, not merely its pairwise version. *Computational evidence.* Use five real vertices $0, \ldots, 4$, dummy 5, and the T_0 quotient with bit order $12, 13, 14, 23, 24, 34$. One first-seat attacker has the spine

$$A : O_0, \quad D : O_2, \quad A : C_0, \quad D : O_1, \quad A : O_5, \quad D : C_2.$$

The selected child has $U = \{3,4\}$, $Q = (1,5)$, quotient score 1, and an attacker continuation with terminal image

$$X = \{1,3\}, \qquad \{0\} \cup (X + X) = \{0,2\}.$$

For example, the two compatible continuations

$$O_4O_3C_1C_5C_4C_3, \qquad O_4C_1O_3C_5C_4C_3$$

have labels 1 and 3. At each of the three preceding defender nodes, every sibling subtree admits an independently chosen deterministic attacker continuation whose entire image avoids $\{0,2\}$. These branchwise continuations combine into one strategy because the first divergent defender history separates their subtrees. The target recursion and all ten sibling checks are executable in `experiments/linking_game.py::quotient_target_response_labels`. Consequently there need not be any single ancestor sibling label in $\{0\}\cup(X+X)$. This still is not a counterexample to affine contraction: several sibling images may have a larger odd dependence even though none contains the proposed one-step escape label.

Proposition 11 proves that even the natural fan over *all* second openings is too rigid. The global affine target survives inside either singleton real branch, so a proof must choose its odd subset of defender branches using continuation data; neither dummy-edge contraction nor the full root fan is a valid fixed recipe.

12 Relation to Gold–Arf normal play

The reductions in `writeups/goldarf.tex` identify the terminal σ-charge with the overlap parity of G-edges. Since overlap and disjointness partition $E(G)$,

$$\sigma(h) = |E(G)| + F_G(h) \pmod 2.$$

Therefore an affirmative answer to Question 40 would force $\sigma = |E(G)|$ for every support graph and both seats.

This arbitrary-graph theorem is not load-bearing for the Gold problem. The deterministic Witt re-framing reduces every loaded support graph to a matching, where the FIFO theorem is proved, and the phase-aware claim compiler turns its forced charge into normal play. Question 40 is a strict combinatorial generalization of the FIFO mechanism rather than a gap in the Gold theorem.

13 Conclusion and evidence boundary

The exact reductions in this paper replace the FIFO game by a causal affine response problem over the binary edge space. They prove the interval and live-star identities, the affine separation criterion, the homogeneous-cell recursion, the cut-space and two-graph decompositions, the correlated cut and continuation moment, the initial-cut contraction, the stopped close-first theorem, and the obstruction results for the stronger local induction templates considered above. The stopped close-first argument and its finite-fan absorbers are also kernel-checked in `formal/Ogdoad/Fifo.lean`.

The remaining statement is Conjecture 40: a causal affine contraction for an arbitrary conditioned tail. Its difficulty is genuinely multibranch. The response factor must transport four-cycle defects across a complete defender fan while allowing odd wall fans, front-cell deletion, matching-phase pivots, and frozen ko-wall edges. The dual descent reduces a minimal counterexample to an even-$|U|$ singleton phase or an imbalanced multicell front, but neither ancestry type is yet contracted.

Exact minimax provides a finite check, not a proof: the conjecture holds on every graph isomorphism class through eight real vertices with an isolated dummy, for both defender seats. The same census confirms the ranked zero-normalized repair forests and the stated root selectors. The explicit counterexamples in this paper show why this evidence does not support the stronger childwise-coset, bounded-support, static-selector, ordered-colour, open-completion, or graph-independent carrier claims. Thus the theorem frontier is narrow and exact: construct the causal factor extension, or produce an arbitrary-order obstruction to it.

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